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(54) **METHODS FOR IMPROVING HEAVY OILS**

(71) Applicant: **PETRORAZA SAS**, Antioquia (CO)

(72) Inventor: **Jose Edgar Patino**, Antioquia (CO)

(73) Assignee: **PETRORAZA SAS**, Antioquia (CO)

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(58) **Field of Classification Search**

CPC . C10G 11/04; C10G 2300/4037; E21B 43/24; E21B 43/205

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,923,130 A 2/1960 Hogan
3,124,418 A 3/1964 Malley et al.
3,794,599 A 2/1974 Dautzenberg et al.
4,460,704 A 7/1984 Twigg
4,640,765 A 2/1987 Kubo
4,720,477 A 1/1988 Hettinger, Jr.
4,724,227 A 2/1988 Kukes et al.
5,146,013 A 9/1992 Dogimont et al.
5,827,489 A 10/1998 Garcin et al.
6,200,927 B1 3/2001 Shukis et al.
7,262,225 B2 8/2007 Van Berge et al.
7,323,100 B2 1/2008 Espinoza et al.
7,426,959 B2 9/2008 Wang
7,449,496 B2 11/2008 Jin et al.
7,563,742 B2 7/2009 Reyes et al.
7,601,668 B2 10/2009 Zhou et al.
7,665,137 B1 2/2010 Zhou et al.
7,712,528 B2 5/2010 Langdon et al.
7,825,064 B2 11/2010 Wong et al.
7,909,097 B2* 3/2011 Ayasse E21B 43/243
166/371

7,981,275 B2 7/2011 Toledo Antonio et al.
9,004,164 B2 4/2015 Dreher
9,169,448 B2 10/2015 Mazyar
9,670,760 B2 6/2017 Ovalles
2004/0127352 A1 7/2004 Jin et al.
2005/0272611 A1 12/2005 Lord et al.
2006/0025301 A1 2/2006 Reddy et al.
2008/0154074 A1 6/2008 Thomazeau et al.
2008/0217008 A1* 9/2008 Langdon E21B 43/2405
166/270
2010/0212893 A1 8/2010 Araghi
2010/0218993 A1 9/2010 Wideman et al.
2011/0000824 A1 1/2011 Zhan et al.
2011/0144382 A1 6/2011 Yin et al.
2011/0152550 A1 6/2011 Grey et al.
2011/0172417 A1 7/2011 Park et al.
2011/0220553 A1 9/2011 Lott et al.
2011/0306490 A1 12/2011 Bhattacharyya et al.
2012/0083643 A1 4/2012 Rashidi et al.
2013/0165353 A1 6/2013 Mazyar et al.
2013/0334097 A1 12/2013 Patino
2015/0175876 A1 6/2015 Resasco et al.
2016/0024374 A1* 1/2016 Sadana E21B 43/164
166/272.6

FOREIGN PATENT DOCUMENTS

EP 0075314 3/1983
EP 1782884 5/2007
JP 2071839 3/1990

OTHER PUBLICATIONS

Kokal, S.L. et al. "Measurement and correlation of asphaltene precipitation from heavy oils by gas injection" Journal of Canadian Petroleum Technology, 1992, pp. 24-30, vol. 31, No. 24.

On, D. Trong, et al., "Perspective on catalytic applications of mesostructured materials", Applied Catalysis A: General, 2003, pp. 545-602, vol. 253, Elsevier Science B.V.

Office Action issued in related Mexican Patent Application No. MX/a/2014/014488 on Sep. 6, 2016, 8 pages with translation.

International Preliminary Report on Patentability of PCT/IB2013/054619, dated Dec. 9, 2014, 16 pages.

International Search Report of PCT/IB2013/054619, dated Jan. 16, 2014, 6 pages.

Written Opinion of PCT/IB2013/054619, 15 pages.

Mexican Office Action of Mexican Patent Application No. MX/a/2014/014488 on Sep. 6, 2016, 4 pages.

Translation of Office Action issued in related Mexican Patent Application No. MX/a/2014/014488 on Sep. 6, 2016, 5 pages.

Partial International Search Report mailed Oct. 29, 2013 for PCT/IB2013/054619 filed Jun. 5, 2013.

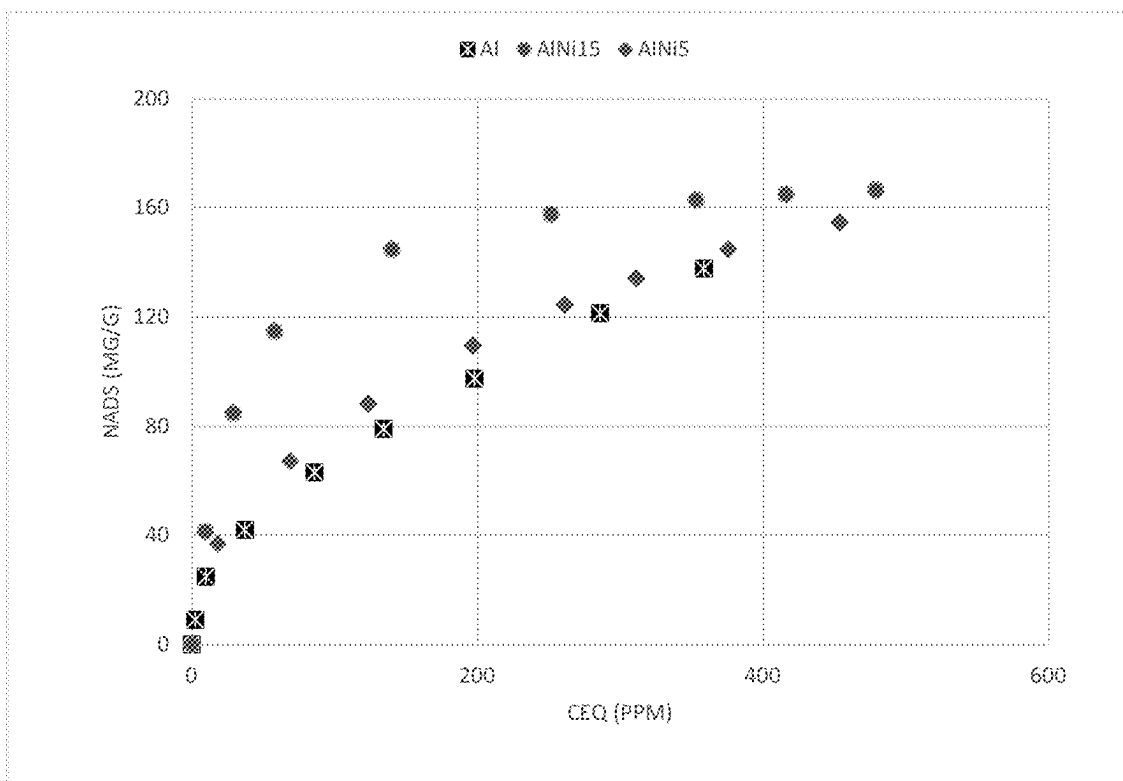
* cited by examiner

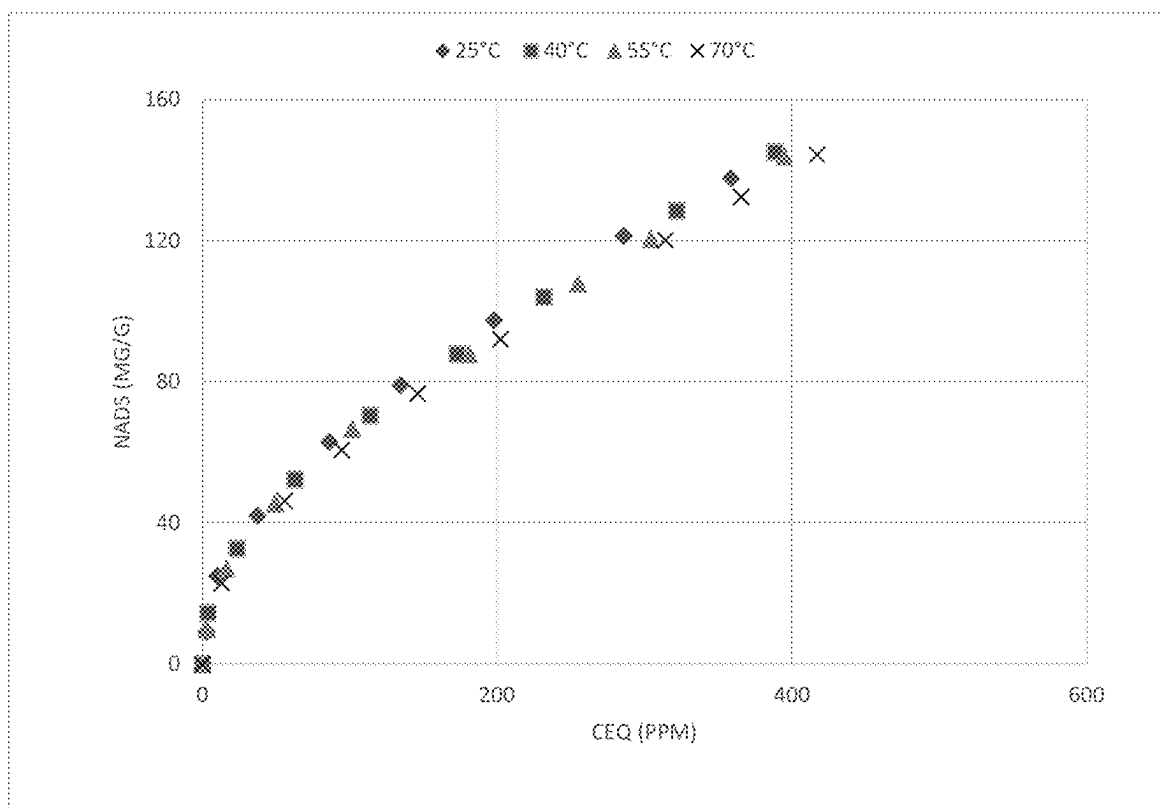
Primary Examiner — Silvana C Runyan

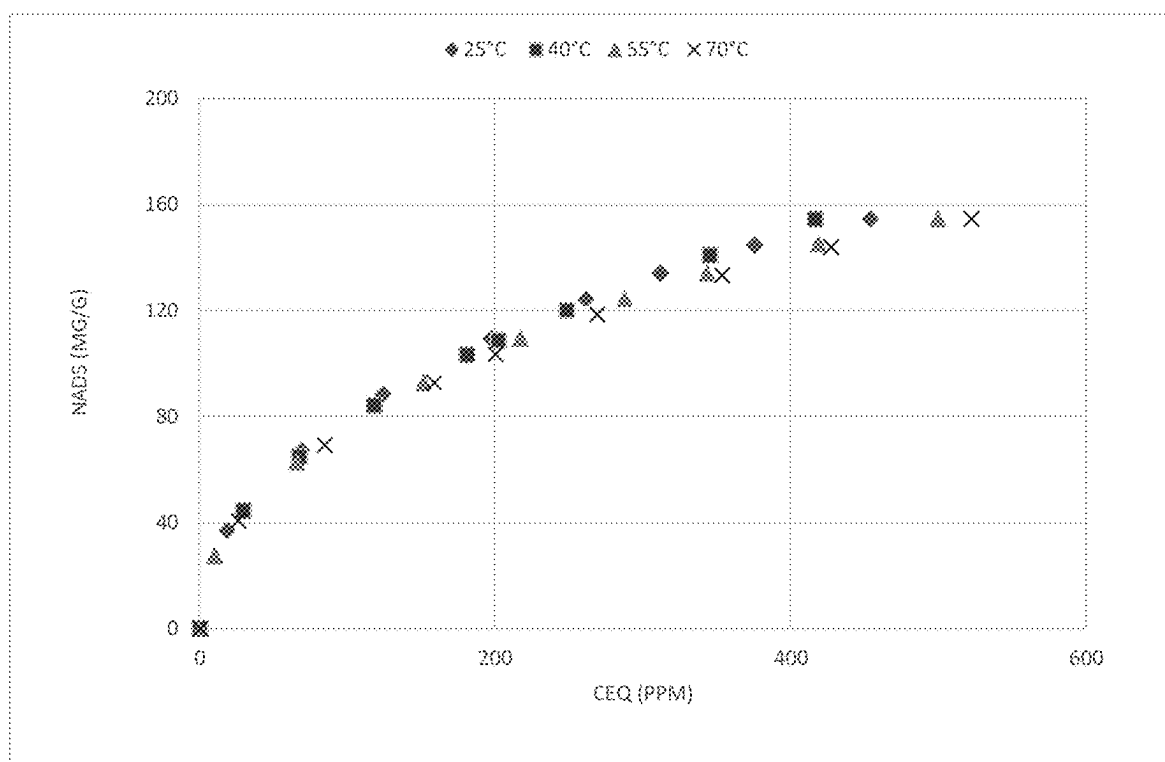
(57) **ABSTRACT**

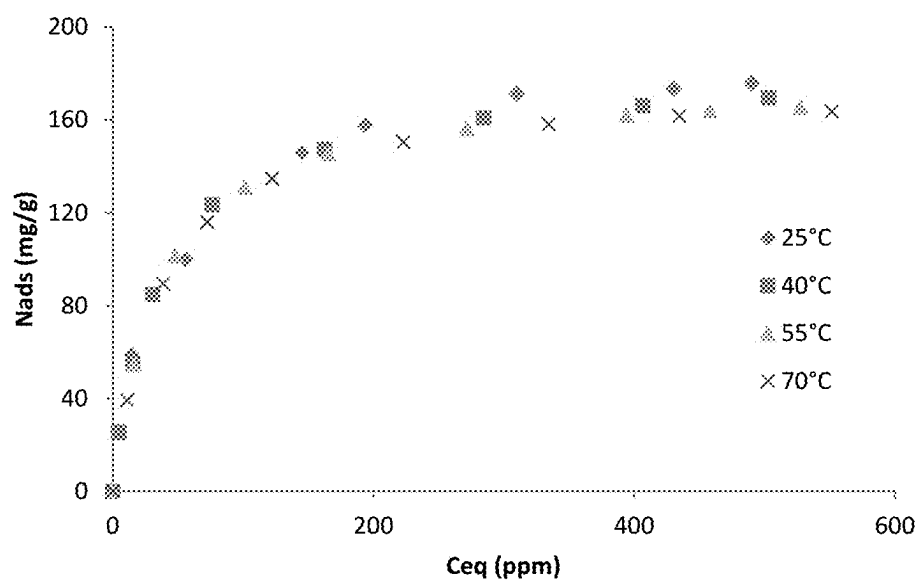
Novel catalysts comprising nickel oxide nanoparticles supported on alumina nanoparticles, methods of their manufacture, heavy oil compositions contacted by these nanocatalysts and methods of their use are disclosed. The novel nanocatalysts are useful, inter alia, in the upgrading of heavy oil fractions or as aids in oil recovery from well reservoirs.

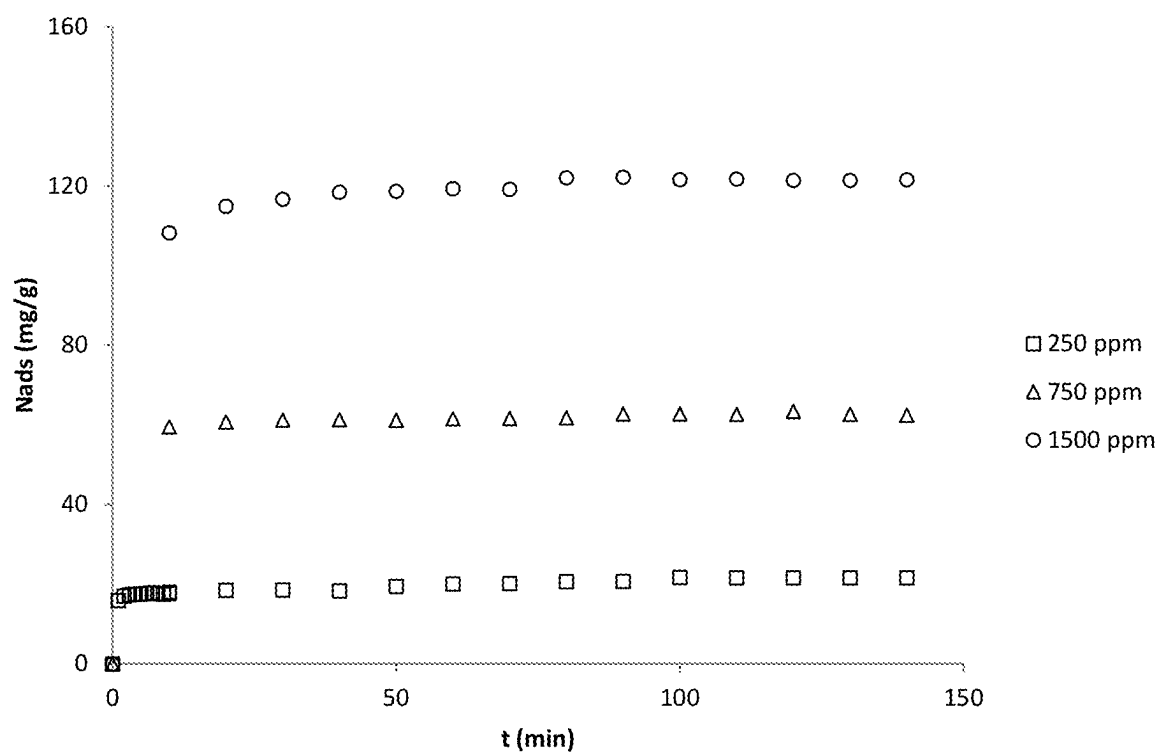
16 Claims, 10 Drawing Sheets

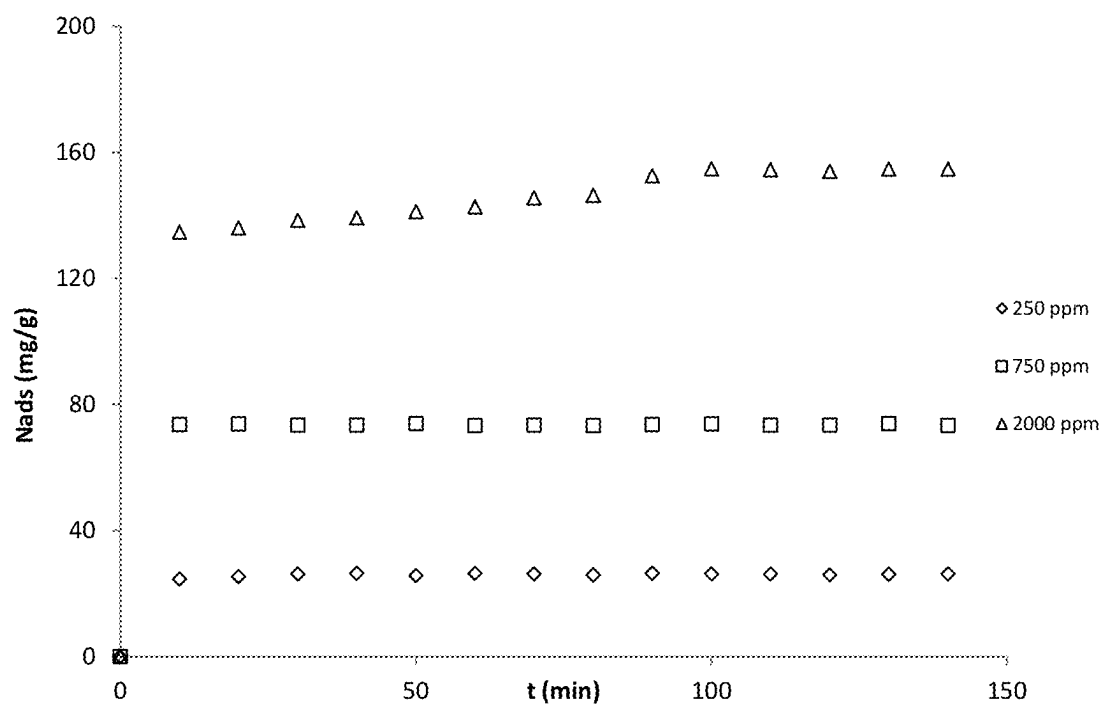
**Figure 1**

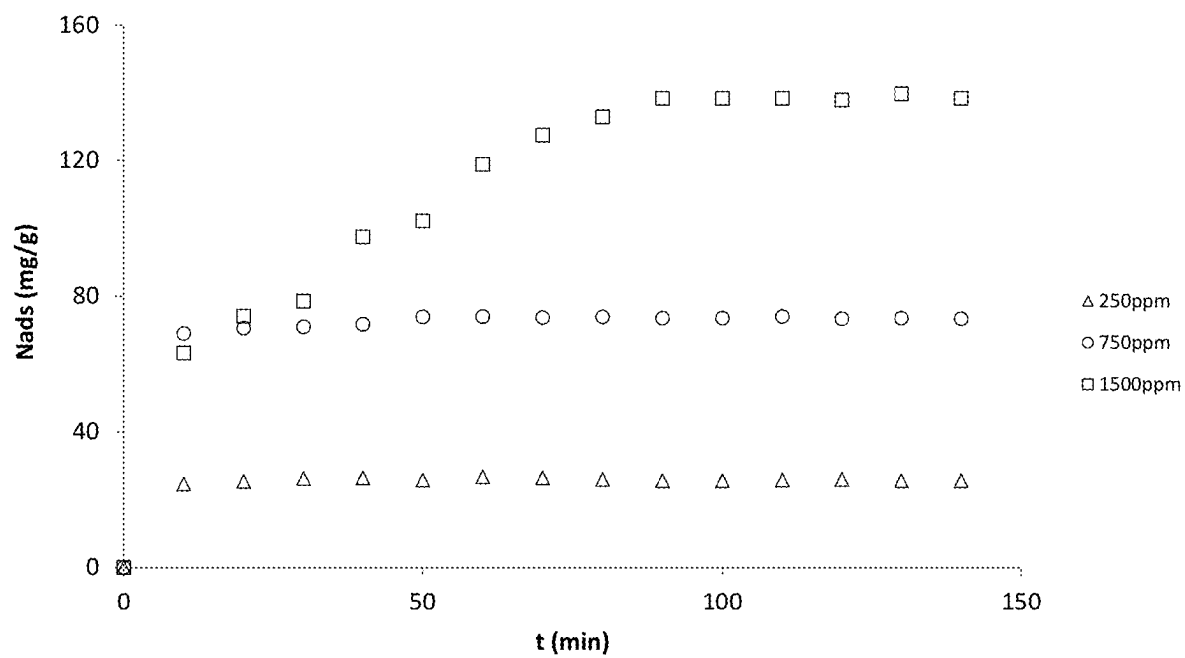
**Figure 2**

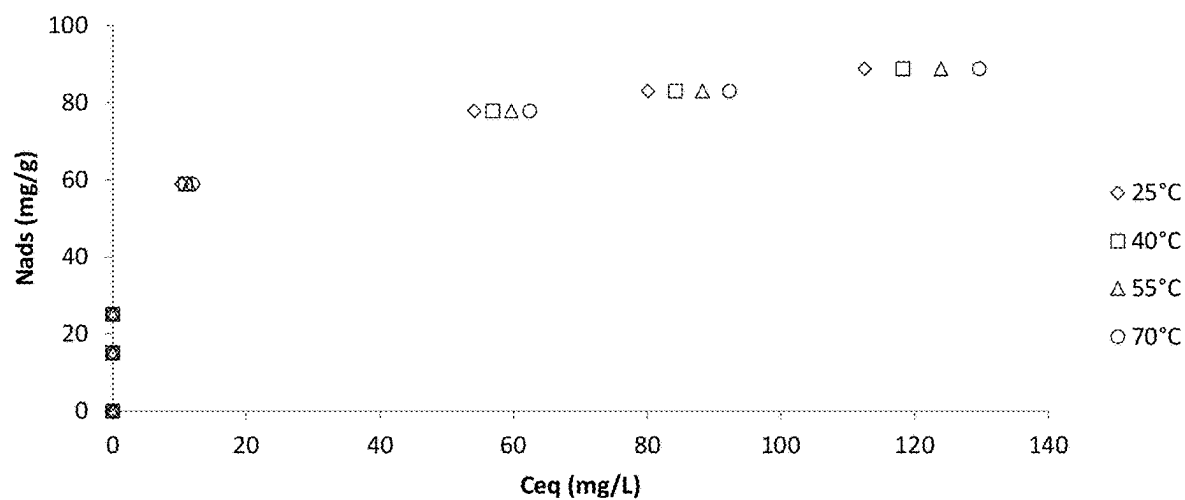
**Figure 3**

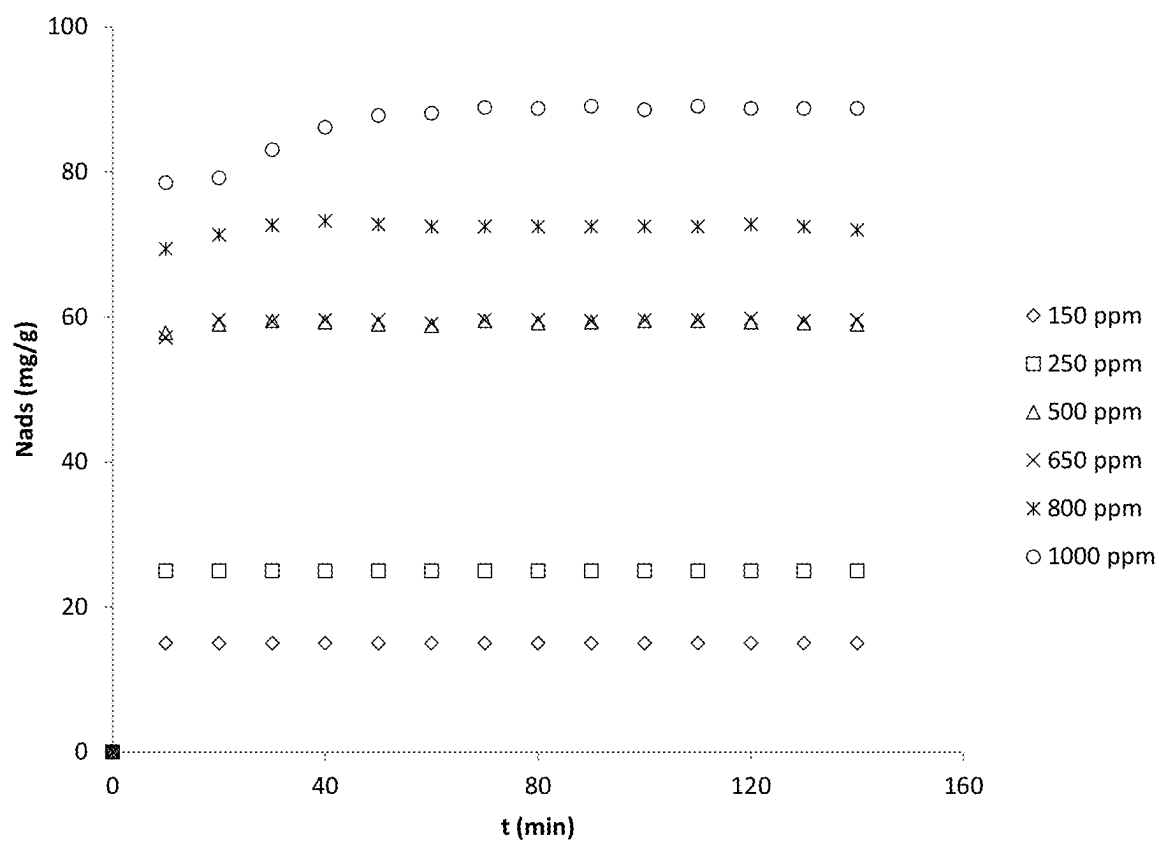
**Figure 4**

**Figure 5**

**Figure 6**

**Figure 7**

**Figure 8**

**Figure 9**

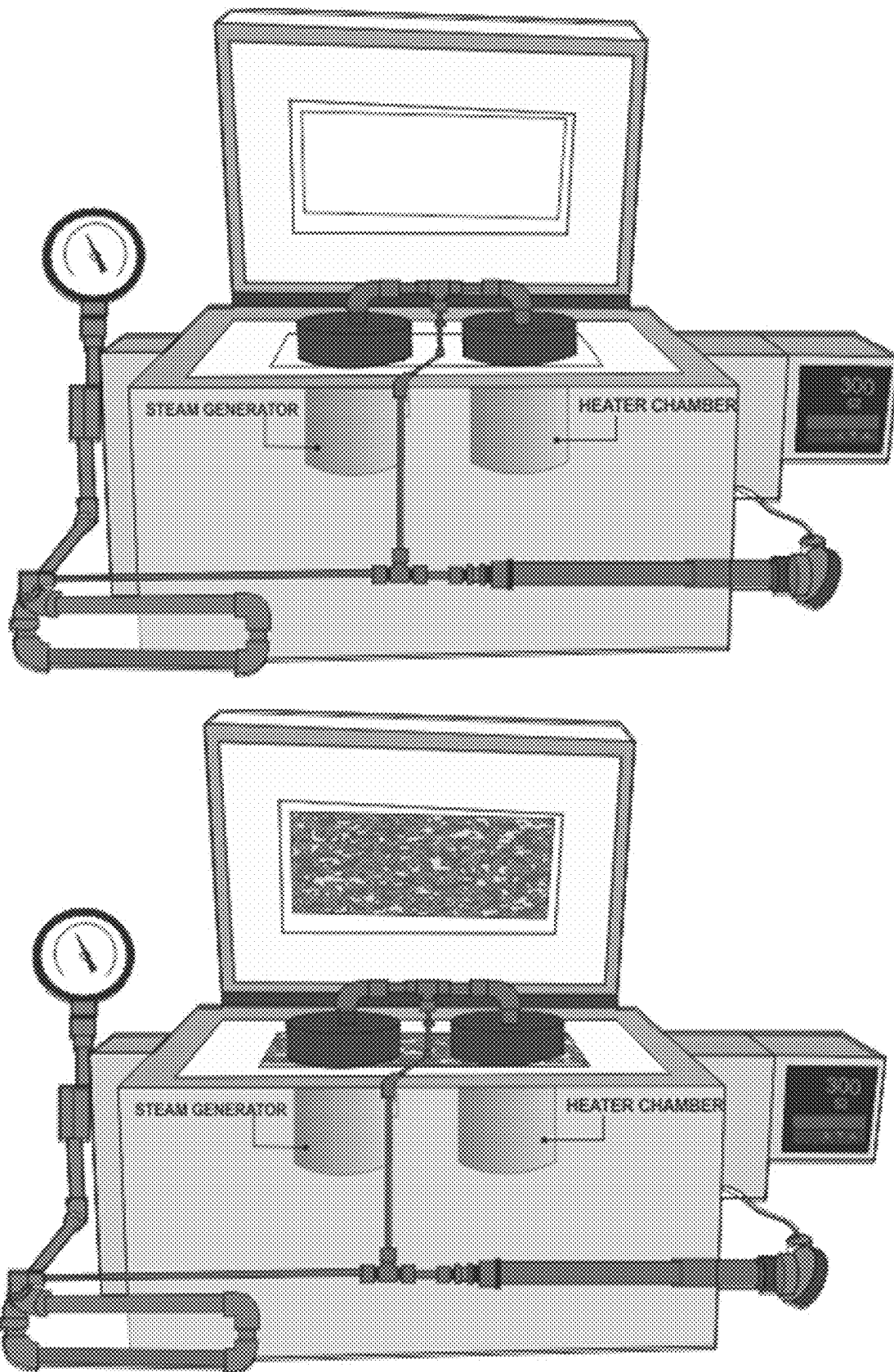


Figure 10

METHODS FOR IMPROVING HEAVY OILS

This application is a continuation of pending U.S. application Ser. No. 18/116,507, filed Mar. 2, 2023, now U.S. Pat. No. 12,006,477, issued Jun. 11, 2024, which is a continuation of U.S. Pat. No. 11,629,295, issued Apr. 18, 2023, which is a continuation of U.S. Pat. No. 11,319,495 issued May 3, 2022, which is a continuation of U.S. Pat. No. 10,907,105 issued Feb. 2, 2021, which is a continuation of U.S. Pat. No. 10,308,879 issued Jun. 4, 2019, which is a continuation of U.S. Pat. No. 10,087,375, issued Oct. 2, 2018, which claims the benefit of U.S. Provisional Application Ser. No. 62/334,180, filed May 10, 2016, the disclosure of each of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates to a new class of supported nanocatalysts that upgrade heavy oil fractions, processes for their preparation, and methods of their use, and products prepared by contacting heavy oil, extra heavy oil, compositions containing heavy oil, extra heavy oil or a combination thereof, including compositions found in geological formations with the nanocatalysts. More particularly, this invention relates to catalysts comprising nickel oxide nanoparticles supported on alumina nanoparticles with improved asphaltene sorption properties that enhance thermal catalytic cracking of heavy oils and/or extra heavy oils within an oil reservoir or during downstream processing.

BACKGROUND OF THE INVENTION

As industrialization expands globally to include an ever enlarging list of countries, demand for oil as an energy source and as feedstock for the myriad of petroleum based products enjoyed by consumers necessarily increases. This demand puts pressure on high quality or readily obtainable oil supplies, and can result in shortages and cost increases. While additional lower quality oil reserves such as heavy oils and bitumen are in abundant supply in Canada, Venezuela and the United States, for example, they generally contain higher levels of high boiling components and/or higher concentrations of impurities such as sulfur, nitrogen or metals. The high boiling fractions typically have a high molecular weight and/or low hydrogen/carbon ratio, an example of which is a class of complex compounds collectively referred to as "asphaltenes". Asphaltenes are difficult to process and commonly cause fouling of conventional catalysts and hydroprocessing equipment. These lower quality feedstocks are further characterized as including relatively high quantities of hydrocarbons that have a boiling point of 524 C (975° F.) or higher. They are typically less attractive to oil producers because they require more expensive processing to break down the high boilers or remove or reduce impurities to acceptable commercial levels that would allow them to effectively compete with light crude. Other examples of lower quality feedstocks that contain relatively high concentrations of asphaltenes, sulfur, nitrogen and metals include bottom of the barrel and residuum left over from conventional refinery processes (collectively "heavy oil").

Shortages and/or price increases in high quality oils help to level the playing field and compensate for any increased costs of heavy oil processing, permitting lower quality oil reserves to become attractive alternatives to light crude. To better compare to light crude, a refiner must modify a

number of properties in heavy oils. In contrast to high quality oils, heavy oils and bitumen are typically characterized by having low specific gravities (0-18 degree. API), high viscosities (>100,000 cp), and high sulfur content (e.g., >5% by weight). Converting heavy oil into useful end products requires extensive processing, including reducing the boiling point of the heavy oil, increasing the hydrogen-to-carbon ratio, and/or removing impurities such as metals, sulfur, nitrogen and high carbon forming compounds. Langdon et al. (U.S. Pat. No. 7,712,528) describes certain heavy oil processing methods generally as well as identifies their shortcomings and the impact of high concentrations of asphaltenes on processing efficiencies.

Other processes reported to hydrocrack heavy oils include those disclosed by Lott et al. (U.S. Published Application No. 20110220553) that is said to disclose methods and systems for hydrocracking a heavy oil feedstock using an in situ colloidal or molecular catalyst. The invention reportedly involves methods and systems for hydroprocessing heavy oil feedstocks that include a significant quantity of asphaltenes and fractions boiling above 524 C (975° F.) to yield lower boiling, higher quality materials and relate to ebulliated bed hydroprocessing methods and systems that employ a colloidal or molecular catalyst and a porous supported catalyst.

To generally reduce the viscosity of oil, the industry has relied on various thermal and catalytic cracking processes. Pyrolysis, or "thermal cracking", typically occurs when oil cracks at temperatures greater than about 650° F. Pyrolysis tends to improve certain heavy oil properties by reducing viscosity and API gravity but may also lead to increased content of acids. By its very nature, thermal cracking generally has minimal effect on total sulfur content. The result is a feedstock that is intrinsically less valuable to downstream processors. Moreover, the high temperatures required increase the likelihood of coke formation, which leads to fouling of refinery equipment or catalysts used by refiners to further process the oil into saleable products. Commercial solutions to these problems include carbon removal or hydrogenation, but costs for these processes must be borne by the refiners. A number of catalysts, including supported nickel catalysts, are available to hydrogenate or hydrotreat oils, but they are typically used in downstream processing. Improvements in nickel-based catalysts may lead to improved efficiencies in these downstream processes, thereby reducing costs and/or increasing product output. A number of processes to prepare certain supported catalysts for use in hydrotreating or hydrogenating various oils are known.

For example, one technique commonly used to obtain supported nickel catalysts starts with the nickel atoms dissolved in a solvent. The nickel atoms are usually provided as nickel salts due to the solubility of nickel salts in various solvents. The support material is added to the nickel solution and the nickel is then precipitated onto the support, typically by adding a base. The supported nickel catalyst is then dried and calcined (e.g., at 375° C.) and activated by reduction with hydrogen.

It is known in the art that heating and/or calcining the catalyst atoms causes agglomeration of catalyst particles to some degree. See Reyes et al., (U.S. Pat. No. 7,563,742). Agglomeration is undesired because it reduces the performance of the catalyst. Agglomerated particles have less exposed surface area and are consequently less active for a given amount of metal (i.e., only the exposed metal atoms on the surface are available for catalysis). Despite the undesirability of agglomeration, exposing the catalyst to heat is

often necessary to activate the catalyst or for carrying out the reactions that involve the catalyst.

The extent of agglomeration during manufacture or use of the catalyst typically depends on the size and number of catalyst particles. Smaller particles are more likely to agglomerate because of higher surface tension as compared to larger particles. Higher metal loading also tends to facilitate agglomeration because the particles are in closer proximity. Although catalyst performance can in theory be increased with smaller catalyst particles, improvement in catalyst performance has been somewhat limited by the inability to beneficially increase metal loading while using small catalyst particles.

Reyes et al. (U.S. Pat. No. 7,563,742) discloses certain supported nickel nanocatalysts having high nickel loadings and methods for their preparation. These catalysts are reportedly useful, inter alia, for hydrocracking, hydrodesulfurization and other similar processes carried out in refinery settings.

Langdon et al. (U.S. Pat. No. 7,712,528) discloses some methods for dispersing nanocatalysts into petroleum bearing formations, forming lighter oil products within the formation, and extracting the lighter oil components from the formation. Processes for the in situ conversion and recovery of heavy crude oils and natural bitumens from subsurface formations are described therein.

Toledo Antonio, et al. (U.S. Pat. No. 7,981,275) reports certain catalytic compositions having a high specific activity in reactions involving hydroprocessing of light and intermediate petroleum fractions, and preferably in hydrodesulfurization and hydrodenitrogenation reactions, employing a catalyst containing at least one element of a non-noble metal from group VIII, at least one element from group VIB and, optionally, a group one element of the VA group, which are deposited on a catalytic support comprising of an inorganic metal oxide from group IVB.

Wong, et al. (U.S. Pat. No. 7,825,064) describes some catalytic materials, and more particularly, catalysts composed of metal oxide on which is supported another metal oxide wherein the support comprises nanometer-sized metal oxide particles.

Espinoza, et al. (U.S. Pat. No. 7,323,100) discloses certain combination of amorphous materials for use in hydrocracking catalysts.

Park, et al. (Published US Application No. 2011/0172417) describes some heterogeneous copper nanocatalysts and methods of their preparation composed of copper nanoparticles on boehmite.

Bhattacharyya, et al. (Published US Application No. 2011/0306490) discloses certain compositions of supported molybdenum catalyst for converting heavy hydrocarbon feed into lighter hydrocarbon products. The support reported is boehmite or pseudo-boehmite and may further contain iron oxide.

Supported catalysts, especially nanocatalysts that maintain or improve catalytic cracking efficiency while requiring lower metal loadings, remain desirable yet elusive targets of the industry. Alternatives employing catalysts that could combine easier recovery of heavy oils from oil bearing formations and improve oil properties would be attractive to oil suppliers and refiners alike. Catalysts and methods of their use for hydroprocessing heavy oil feedstocks that include a significant quantity of asphaltenes and fractions boiling above 570° C. (1,058° F.) to yield lower boiling, higher quality materials are also desirable. Catalysts and methods of their use that, by their use in treat heavy oils in formation and recovery, extend the useful life of expensive

equipment used to extract or further process the upgraded heavy oil fractions would be of commercial interest. The invention is related to these and other important ends.

SUMMARY OF THE INVENTION

Accordingly, the present invention is directed, in part, to catalysts comprising: nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500.

In other embodiments, the present invention is directed to processes for preparing a catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500;

said process comprising:

dry impregnating an amorphous dried sodium aluminate precipitate with an aqueous solution of a water-soluble nickel salt; and

drying the nickel impregnated precipitate;

wherein the dry impregnating and drying steps are each carried out for a time and under conditions sufficient to provide the nickel impregnated precipitate catalyst.

In yet other embodiments, the present invention is directed to catalysts prepared by the processes for preparing a catalyst described herein.

In still other embodiments, the present invention is directed to methods for upgrading heavy oil fractions in a well, comprising:

contacting the heavy oil in a well producing heavy oil with a catalyst according to the invention for a time and under conditions sufficient to increase the H/C ratio.

In yet other embodiments, the present invention is directed to upgraded heavy oil fractions prepared by the processes for upgrading heavy oil fractions described herein.

The foregoing and other objectives, features, and advantages of the invention will be more readily understood upon consideration of the following detailed description of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows the asphaltene sorption isotherms at 25° C. for an alumina nanoparticle support as disclosed and certain alumina nanoparticle catalysts according to the present invention having different nickel oxide nanoparticle loadings.

FIG. 2 shows the asphaltene sorption isotherm at a variety of temperatures for an alumina nanoparticle support as disclosed.

FIG. 3 shows the asphaltene sorption isotherm at a variety of temperatures for an alumina nanoparticle catalyst according to the present invention having a nickel oxide nanoparticle loading (AlNi (5%)).

FIG. 4 shows the asphaltene sorption isotherm at a variety of temperatures for an alumina nanoparticle catalyst according to the present invention having a nickel oxide nanoparticle loading (AlNi (15%)).

FIG. 5 shows the amount of asphaltene adsorbed on alumina versus time for different initial concentrations of asphaltenes.

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FIG. 6 shows the amount of asphaltene adsorbed on AlNi (5%) versus time for different initial concentrations of asphaltenes.

FIG. 7 shows the amount of asphaltene adsorbed on AlNi (15%) versus time for different initial concentrations of asphaltenes.

FIG. 8 shows the asphaltene sorption isotherm at a variety of temperatures for an alumina nanoparticle catalyst according to the present invention having nickel oxide and Palladium nanoparticle loadings (Pd0.5/Ni5/Al).

FIG. 9 shows the amount of asphaltene adsorbed on Pd0.5/Ni5/Al versus time for different initial concentrations of asphaltenes.

FIG. 10 is an exemplary design of an apparatus used to carry out laboratory protocols used to mimic conditions in a steam assisted well for evaluating certain of the methods disclosed in this application.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

As employed above and throughout the disclosure of the present invention, the following terms, unless otherwise indicated, shall be understood to have the following meanings.

As used herein, the term “activated alumina” refers to materials manufactured from aluminum hydroxide by dehydroxylating it in a way that produces a highly porous material; which can have a surface area significantly over 200 square meters/gram.

As used herein the term “refractory material” refers to a material that retains its strength at high temperatures. ASTM C71 defines refractories as “non-metallic materials having those chemical and physical properties that make them applicable for structures, or as components of systems, that are exposed to environments above 1,000° F. Refractory materials must be chemically and physically stable at high temperatures. Depending on the operating environment, they need to be resistant to thermal shock, be chemically inert, and/or have specific ranges of thermal conductivity and of the coefficient of thermal expansion. The oxides of aluminum (alumina), silicon (silica) and magnesium (magnesia) are the most important materials used in the manufacturing of refractories. Another oxide usually found in refractories is the oxide of calcium (lime). Fire clays are also widely used in the manufacture of refractories. Refractories must be chosen according to the conditions they will face. Some applications require special refractory materials. Zirconia is used when the material must withstand extremely high temperatures. Silicon carbide and carbon (graphite) are two other refractory materials used in some very severe temperature conditions, but they cannot be used in contact with oxygen, as they will oxidize and burn.

As used herein, the term “nanoparticle” refers to fine particles having a particle size of less than or equal to 100 nanometers (i.e., less than or equal to 0.1 μm)

As used herein, “about” will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which it is used. If there are uses of the term which are not clear to persons of ordinary skill in the art given the context in which it is used, “about” will mean up to plus or minus 10% of the particular term.

This invention is directed to, inter alia, the surprising and unexpected discovery of a new class of supported nanocatalysts that upgrade heavy oil fractions, processes for their preparation, and methods of their use, and products prepared by contacting heavy oil fractions with the nanocatalysts.

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More particularly, this invention relates to catalysts comprising nickel oxide nanoparticles supported on alumina nanoparticles with improved asphaltene sorption properties that enhance thermal catalytic cracking of heavy oils within an oil reservoir or during downstream processing.

Among the methods that may be employed for measuring nickel oxide (NiO) on the final catalyst are hydrogen adsorption, atomic absorption and certain gravimetric method. However, hydrogen adsorption is the most accurate method because it identifies active sites on the surface of the support material which are the sites having NiO content. Atomic absorption spectrometry (AAS): is the well-known methodology in which radiant energy is emitted from a hollow cathode lamp and passed through a flame. Each atomic element energy band is very narrow and is easily distinguished from the other atomic absorption lines of other elements. Gravimetric methods are very simple in their nature and much less accurate. They estimate the difference in weight before and after impregnation of dry material.

The term “asphaltenes” as used herein refers to the fraction of oil, bitumen or vacuum residue that is insoluble in low molecular weight paraffins such as n-heptane or n-pentane, while being soluble in light aromatic hydrocarbons such as toluene, pyridine or benzene.

Benefits of the catalysts include one or more of the following: improved catalyst tolerance for impurities found in heavy oils; ease of upgrading or hydrocracking the heavy oils, preferably wherein the cracking is carried in the well reservoirs, said oils having increased levels of hydrocracked materials contained therein after contacting with the catalysts and hydrogen transfer agents; reduced levels of impurities such as sulfur; lower reaction temperatures and/or reaction pressures for carrying out the in-situ reactions in the hydrocarbons reservoirs; limited downtimes for wells while the hydrocracking takes place and the like. In certain of the methods employing the catalysts of the invention for use in upgrading heavy oils, the nanocatalyst does not require heat in addition to that experienced in the underground reservoir. Preferably, the catalyst suitably upgrades the heavy oil in a well at temperatures of no higher than 250° F., more preferably at temperatures below 250° F., still more preferably below 240, 220, 200, or even 180° F.

Accordingly, in certain embodiments, the present invention provides catalysts comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400.

In certain preferred embodiments of the catalysts according to the invention, the nickel oxide (NiO) nanoparticles are present in an amount of about 0.2% to about 1% by weight of catalyst, preferably as measured by hydrogen adsorption.

In other preferred embodiments, the particle size of the nickel oxide nanoparticles or the alumina nanoparticles comprising the catalysts is less than about 0.1 μm , more preferably wherein the particle size of the nickel oxide nanoparticles and the alumina nanoparticles are each less than about 0.1 μm .

In some preferred embodiments, the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

In yet other preferred embodiments, the catalysts further comprise nanoparticles of at least one Group VIIIB metal oxide supported on the alumina nanoparticles;

wherein:

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the Group VIIIB metal is other than nickel, preferably selected from platinum, palladium, and iron, and combinations thereof; and

the alumina nanoparticle to Group VIIIB metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400.

In certain other preferred embodiments, the catalysts further comprise nanoparticles of at least one Group IB metal supported on the alumina nanoparticles;

wherein:

the alumina nanoparticle to Group IB metal nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400; and wherein the Group IB metal is preferably silver.

In some embodiments, the invention is directed to processes for preparing a catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400;

said process comprising:

dry impregnating an amorphous dried sodium aluminate precipitate with an aqueous solution of a water-soluble nickel salt; and

drying the nickel impregnated precipitate; wherein the dry impregnating and drying steps are each carried out for a time and under conditions sufficient to provide the dried nickel impregnated precipitate catalyst.

In certain preferred embodiments, the catalyst or catalyst intermediate is a dried nickel impregnated precipitate prepared by a process disclosed herein.

In certain preferred embodiments, the processes further comprise calcining the dried nickel impregnated precipitate in the presence of oxygen or air;

wherein:

the calcining is carried out for a time and under conditions sufficient to provide the calcined nickel catalyst.

In certain more preferred embodiments, the invention is directed to a calcined dried nickel impregnated precipitate catalyst prepared by a process disclosed herein.

In some other embodiments, the invention is directed to processes for preparing a calcined catalyst, said catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400; and

nanoparticles of at least one Group VIIIB metal oxide nanoparticles supported on the alumina nanoparticles; wherein:

the Group VIIIB metal is other than nickel, preferably palladium, platinum or iron, and combinations thereof; and

the alumina nanoparticle to Group VIIIB metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400;

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said process comprising

dry impregnating an amorphous dried sodium aluminate precipitate with an aqueous solution of a water-soluble nickel salt; and

drying the nickel impregnated precipitate;

wherein the dry impregnating and drying steps are each carried out for a time and under conditions sufficient to provide the dried nickel impregnated precipitate catalyst;

dry impregnating the dried nickel impregnated precipitate with an aqueous solution of a water-soluble Group VIIIB metal salt;

drying the nickel and Group VIIIB metal impregnated precipitate; and

calcining the dried nickel and Group VIIIB metal impregnated precipitate in the presence of oxygen or air;

wherein each of the dry impregnating, drying, and calcining are carried out for a time and under conditions sufficient to provide the calcined nickel and Group VIIIB metal impregnated catalyst.

In certain more preferred embodiments, the invention is directed to a calcined dried nickel and Group VIIIB metal impregnated precipitate catalyst prepared by a process disclosed herein.

In some other embodiments, the invention is directed to processes for preparing a calcined catalyst, said catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400; and

nanoparticles of at least one Group IB metal oxide nanoparticles supported on the alumina nanoparticles; wherein:

the Group IB metal is preferably silver; and

the alumina nanoparticle to Group IB metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500, preferably in a range of from about 99 to about 400;

said process comprising

dry impregnating an amorphous dried sodium aluminate precipitate with an aqueous solution of a water-soluble nickel salt; and

drying the nickel impregnated precipitate.

wherein the dry impregnating and drying steps are each carried out for a time and under conditions sufficient to provide the dried nickel impregnated precipitate catalyst;

dry impregnating the dried nickel impregnated precipitate with an aqueous solution of a water-soluble Group IB metal salt;

drying the nickel and Group IB metal impregnated precipitate; and

calcining the dried nickel and Group IB metal impregnated precipitate in the presence of oxygen or air;

wherein each of the dry impregnating, drying, and calcining are carried out for a time and under conditions sufficient to provide the calcined dried nickel and Group IB metal impregnated catalyst.

In certain more preferred embodiments, the invention is directed to a calcined dried nickel and Group IB metal impregnated precipitate catalyst prepared by the process disclosed hereinabove.

In certain preferred embodiments of the present catalyst compositions, processes and methods, the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst as described herein.

Typically, the alumina nanoparticles are derived from aluminum metal or an aluminum containing compound that has been contacted with an aqueous alkaline material such as hydroxide, preferably potassium or sodium hydroxide, more preferably sodium hydroxide. While any aluminum compound capable of dissolution in aqueous base may be employed, in certain preferred embodiments, aluminum metal is used as the aluminum feedstock. In other alternately preferred embodiments, aluminum hydroxide is used. Once the aluminum or aluminum hydroxide is dissolved, it may be precipitated as an amorphous solid by reacidification by adding an acid and monitoring the pH until it is in the range of from about 8 to about 8.5, preferably about 8.5. Preferably, the reacidification may be accomplished by using gaseous CO₂ bubbled slowly into the solution, more preferably at room temperature. At this point, the acid addition may be terminated and the aluminum precipitate may be isolated, for example, by filtration. The isolated precipitate may be used in the metal impregnation step, preferably by an incipient wetness method of impregnation (also referred to at times as "dry impregnation"), after its drying. In certain instances, it is advantageous to wash the precipitate one or more times with water, preferably distilled or deionized water, after isolation to reduce the level of sodium, potassium or other cation associated with the alkaline material prior to the drying step. Once the aluminate precipitate has been dried, it is ready for the dry impregnation step with any of the identified metal salts. It is not necessary to calcine the aluminum precipitate prior to impregnation. In certain instances it is preferred that the aluminum precipitate not be calcined prior to metal impregnation.

In other preferred embodiments, the dried sodium aluminate precipitate is dry impregnated with an aqueous solution of a water-soluble nickel salt by employing the incipient wetness method (IWM). Preferably, the water-soluble nickel salt comprises nickel nitrate, nickel chloride or nickel sulfate, more preferably nickel nitrate. Typically in the IWM, the active metal precursor is dissolved in an aqueous solution. Then the metal-containing solution is added to a catalyst support containing the same pore volume as the volume of solution that was added. Capillary action draws the solution into the pores. The catalyst can then be dried and calcined to drive off the volatile components within the solution, depositing the metal on the catalyst surface. The maximum loading is limited by the solubility of the precursor in the solution. The concentration profile of the impregnated compound depends in the mass transfer conditions within the pores during impregnation and drying. Alternatively, the precipitate may be prepared by any of the processes known to the ordinarily skilled artisan.

To remove any volatiles following impregnation by the incipient wetness method, the precipitate may be dried by heating for a period of time until the volatiles, such as water are removed. In certain preferred embodiments, the nickel impregnated precipitate is dried at a temperature in the range of from about 100 to about 140° C. for a time sufficient to remove substantially all of the water from the nickel impregnated precipitate, preferably for from about 3 to about 8 hours. These conditions are generally recognized by the skilled artisan as insufficient to calcine the metal impregnated precipitates of the present invention.

In some other preferred embodiments of the processes described herein, the dried nickel impregnated precipitate is

thereafter calcined in the presence of oxygen or air for a time and under conditions sufficient to provide the calcined catalyst. A variety of conditions sufficient to calcine the dried nickel impregnated precipitate are well known to the ordinarily skilled artisan. In certain more preferred embodiments of the present invention, the dried nickel impregnated precipitate is calcined at a temperature in the range of from about 400 to about 500° C. for a time sufficient to calcine the catalyst, preferably for from about 3 to about 8 hours.

In certain alternatively preferred embodiments, the dried nickel impregnated precipitate may then be further impregnated with at least one additional metal salt, preferably one additional metal salt, preferably by the incipient wetness (or dry impregnating method). Preferably the metal further impregnating the nickel impregnated precipitate is a Group VIIIB or Group IB metal salt, preferably dry impregnated, more preferably with a water-soluble Group VIIIB or Group IB metal salt. When the metal is a Group VIIIB metal, it is preferably a water soluble salt of palladium, platinum or iron. When the metal is a Group IB metal, it is preferably a water soluble salt of silver. These water soluble salts may comprise counterions of chloride, sulfate or nitrate, preferably nitrate. To remove any volatiles following impregnation by the incipient wetness method, the multiply metal impregnated precipitate may be dried by heating for a period of time until the volatiles, such as water are removed. In certain preferred embodiments, the multiply metal impregnated precipitate is dried at a temperature in the range of from about 100 to about 140° C. for a time sufficient to remove substantially all of the water from the multiply metal impregnated precipitate, preferably for from about 3 to about 8 hours. Thereafter, the multiply metal impregnated precipitate is preferably calcined in the presence of oxygen or air for a time and under conditions sufficient to provide the calcined catalyst. A variety of conditions sufficient to calcine the dried multiply metal impregnated precipitate or dried nickel impregnated precipitate are well known to the ordinarily skilled artisan. In certain more preferred embodiments of the present invention, the dried nickel impregnated precipitate or dried multiply metal impregnated precipitate is calcined at a temperature in the range of from about 400 to about 500° C., preferably for from about 3 to about 8 hours.

In certain embodiments, the invention is directed to methods for upgrading heavy oil fractions, preferably in a well, comprising:

contacting the heavy oil, preferably in a well that produces heavy oil, with a catalyst according to the invention for a time and under conditions sufficient to increase the H/C ratio.

The H/C ratio of the heavy oil fractions may be measured by any number of methods known to the skilled artisan. In certain preferred embodiments, the H/C ratio is measured by elemental analyzer EXETER CE-490.

The upgrading down well may take place in the following fashion. This general upgrading procedure may be employed after a well has been drilled and completed, whether or not the well is currently in production. Prior to introducing the catalyst into the well, it is useful if the well is perforated within the target zones that contain oil. A volume of treatment fluid containing catalyst is calculated, based on a radial volume of usually 7-20 feet surrounded the well bore in the target zone. This volume is calculated for the effective pore volume based on the rock reservoir porosity. To pump the fluid into the well, it is advantageous to use a coiled tubing that runs through the well head and into position near the front of the perforations of target zone (pay zone) in the reservoir. Then, the fluid containing catalyst is injected or

squeezed into the well by a capillary string or through use of a coiled tube and flows through the perforations into the target zones at a pressure higher than the formation pressure. Other methods are recognized by the skilled artisan. As used herein, the term "coiled tubing" or "coiled tube" refers to a continuous length of steel or composite tubing that is flexible enough to be wound on a large reel for transportation. The coiled tubing unit is typically composed of a reel with the coiled tubing, an injector, control console, power supply and well-control stack. The coiled tubing is injected into the existing production string, unwound from the reel and inserted into the well". Target formations (called pay zones or target zones) absorb the fluid as it is being injected. The pumping rate is set so as not to reach or exceed the formation fracture pressure, a characteristic defined by the geology of the individual well. Once the volume of the fluid has been squeezed into the formation, injection ceases and the well is maintained for a period of time ("soaking") in a static condition (no oil removal) to allow the desired reaction to take place. An exemplary time for "soaking" is overnight. During this time, the catalyst is in contact with the crude oil in the formation at the temperature and pressure that are defined by the well itself. For example, the Chichimene well temperature is about 80° C. and formation pressure is about 1500 PSI. After sufficient time has been allowed for the soaking, the well is reopened and fluids from the target zones (pay Zone) begin to flow back to the surface. In certain preferred embodiments, the well is retreated with additional catalyst after a time, preferably from about a few months after the most recent treatment to about a year, or even more after the most recent treatment with the nanocatalyst of the present invention.

In some preferred embodiments of the methods described herein, the heavy oil fraction contacting with catalyst further comprises contacting with a hydrogen transfer agent. Exemplary hydrogen transfer agents include alcohols or donor solvents, more preferably 1,2,3,4-tetrahydronaphthalene.

In certain more preferred embodiments, the invention is directed to upgraded heavy oil fractions, preferably those produced in a heavy oil producing well, such fractions produced by the methods of the present invention.

Heavy oil and/or extra heavy oil are typically found in formations that are significantly more shallow than light oil formations. Heavy oil formations, such as, for example, oil sands, may be located at from about 200 to about 2000 feet below the earth's surface. At these depths, the temperature of the formation is generally not hot enough to reduce the viscosity of the heavy oil and/or extra heavy oil found in the crude sufficiently to make the oil amenable to pumping to the surface and/or to initiate hydrocracking either of which, alone or together, may decrease overall viscosity of the trapped oil. For example, some heavy oil formations, at least in part due to their shallowness, have a temperature in a range of from about 80° to about 120° F. In order to elevate the temperature of the formation, oil producers may employ steam injection under pressure to heat the well to assist in crude heavy oil removal. To steam heat the formation usually takes an extended period of time, and is carried out at considerable cost to the heavy oil producer. Accordingly, many formations are not reheated when production declines and the easiest oil to remove has been pumped to the surface. Once Steam injection of such wells may be carried out as follows: First, at least one injector well and one or more parallel producer wells are drilled, wherein a section of each of the wells is perforated (typically using narrow slits in the pipe) to allow steam to be delivered from the well into the surrounding heavy oil formation, without permitting the oil

sands in the formation to enter into the perforated wells. The production zones of these wells may be substantially vertical or substantially horizontal, or at any angle in between. For example, steam assisted gravity drainage wells (SAGD) employ substantially horizontal injector wells with substantially parallel producer wells, wherein the producer well is located below and in parallel to the injector well so that the heavy oil drains by gravity into the producer well. The typical construction of SAGD wells is well understood by the person of ordinary skill in the art.

Steam is then injected under slight positive pressure in both injector and producer wells so as to deliver steam, and thusly heat, to the formation, but without fracking. The steam injection is continued by pressuring the wells for a time sufficient to heat the formation to a temperature sufficient to reduce viscosity of the crude oil to an extent where the oil can be pumped to the surface. This heating may be considered to create a heat chamber encompassing a portion of the crude oil in the formation. In many instances this heating/steam injection is continued for a month or more, more typically, 2 to four months, until the desired temperature is achieved in the heat chamber. The desired temperature range in these wells is typically at least about 300° C., to about 350°, 400°, 450°, 500° or even about 550° C. At the higher temperatures, hydrocracking of asphaltenes is usually initiated. However, the failure to attain the higher temperatures necessary for desirable hydrocracking can lead to coking, which may affect pumpability of the heated oil. The steam pressures in the well during the formation heating step are typically those associated with pressure that accommodate the steam temperature required to heat the heat chamber.

Once the desired temperature is achieved, the steam is stopped in the one or more producer wells, which are then employed to transfer the now less viscous crude heavy oil from the formation to the surface. Steam may continue to be injected under pressure into the injection well (artificial pump relief), but typically only to make up any pressure loss created in the formation as heavy oil is removed and pumped to the surface via the producer well(s) and/or maintain the temperature in the heat chamber.

Methods that possess the capability of improving the efficiency of heat transfer from steam to the oil formation, assisting in further reducing viscosity of the formation's contained heavy oil, are able to extract more of the oil from the reservoir (i.e. a heavy oil formation), assist in producing oil from formations presently considered economically unviable, and/or extend the time before a well must be reheated to cost effectively produce this oil from the formation, are of great interest to producers of oil.

Accordingly, in some embodiments, the invention relates to methods for down well viscosity reduction of heavy oil from steam assisted wells containing heavy oil.

In other embodiments, the invention is directed to methods for improving the efficiency of heat transfer in steam assisted wells containing heavy oil from injected steam to the heavy oil formation.

In certain other embodiments, the invention is directed to methods for enhancing oil production for steam assisted heavy oil wells.

In yet other embodiments, the invention is directed to methods for extending the time before a well must be reheated to cost effectively produce heavy oil from the heavy oil formation.

In other embodiments, the invention is directed to methods for improving the quality of the heavy oil contained in the heavy oil formation in-situ.

Typically, the methods of the present invention employ a nanocatalyst in combination with steam injection to prepare the well for heavy oil extraction. In general, the nanocatalyst may be any nanocatalyst having one or more of the properties of: lowering the temperature at which asphaltenes may be hydrocracked and/or improving heat transfer efficiency of the well-injected steam to the heavy oil formation (excellent heat transfer characteristics). The diameter of the nanocatalyst must be of a proportion that it passes through the perforations in the well pipes into the formation. It is also useful for the nanocatalyst and any carrier fluid containing such nanocatalyst to be able to travel (spread) efficiently and effectively into the formation to deliver the catalyst to a greater amount of trapped oil. The nanocatalysts and related carrier fluids as described herein throughout typically contain one or more of these properties and, as such, are usefully employed in any of the methods disclosed herein. The nanocatalysts and carrier fluids are also described in U.S. patent application Ser. No. 13/896,578, recently allowed, hereby incorporated herein by reference in its entirety.

In certain embodiments, the wells to which the present methods are directed are steam-assisted heavy oil wells. In some embodiments, the wells are substantially vertical in nature. In vertical wells where there are two or more producer wells, it is common to place the injector well in the center and have the producer wells disposed about the injector well. In other embodiments, the wells are substantially horizontal in nature. In embodiments wherein the wells are substantially horizontal, they are preferably SAGD heavy oil wells. That is to say, the two wells have producing legs that are essentially horizontal, and parallel to each other, wherein the injector well is above the producer well, such that the heated and more fluid heavy oil flows toward the producer well leg by gravity. While the steam-assisted wells may generally be associated with reservoirs/heavy oil formations at any depth below the earth's surface, the wells are typically at a depth of from about 200 to about 2000 feet below the surface. In substantially horizontal wells, the horizontal leg of the injection or producer well may extend from about 200 to about 3000 feet in the substantially horizontal direction within the heavy oil formation.

While the nanocatalyst and/or carrier fluid may be injected into either the injector well or the producer well, or both, and at levels in any relative ratio one to the other, the nanocatalyst and/or carrier fluid are typically initially steam-injected into the injector and producer wells of interest. The levels of nanocatalyst and/or carrier fluid may be equally disposed between the injector well and the producer well, but it is generally advantageous to put more nanocatalyst and/or carrier fluid into the injector well than into the producer well. Exemplary ratios of nanocatalyst and/or carrier fluid levels placed into the injector well as compared to that injected into the producer well include about 55/45, 60/40, 65/35, 70/30, 75/25, 80/20, 85/15, 90/10 and even about 95/5, and all combinations and subcombinations of ratios between. While not wishing to be held to any theory, it is believed that the nanocatalyst will remain in the formation longer if primarily injected into the injector leg, at least in part because it necessarily must travel further to reach the producer leg and be carried out with the heavy oil being pumped out of the well.

The nanocatalyst and/or carrier fluid weights/volumes are estimated as described herein to permeate a displaceable volume within the heavy oil formation about the well legs being injected, said volume typically having about a 6" to about a 12" radius surrounding the injector leg and/or producer leg(s), and preferably extending at least about 20,

30, 40, 50, 60, 70, 75, 80, 85, 90, 95, and up to about 99% (and all combinations or subcombinations thereof) or more of the perforated section of the injector or producer leg(s).

Generally speaking, the volume of carrier fluid used to inject the nanocatalyst typically contains from about 200 to about 4000 ppm nanocatalyst, preferably from about 500 to about 3000 ppm, more preferably from about 1000 to about 2000 ppm. To minimize costs associated with the addition of the catalyst, the level introduced into the well is typically limited to these levels, although lesser or greater may work as well, depending on the particular well, the make-up of the heavy oil, and/or the amount of oil estimated in the well available for extraction. The nanocatalyst is preferably a nickel on alumina nanocatalyst or a Nickel/Group VIIIb (i.e., Pd or Pt or combination thereof) nanocatalyst, more preferably nanocatalyst compositions as disclosed herein, still more preferably a nickel on alumina nanocatalyst or a Nickel/Group VIIIb (i.e., Pd or Pt or combination thereof) nanocatalyst as disclosed herein. In some embodiments, the Nickel or Nickel/Group VIIIb on alumina nanocatalysts facilitate heat transfer of the steam's heat to the heavy oil formation, thereby improving overall steam efficiency and/or reducing the time required to heat the heat chamber to the desired temperature. This may result in lower costs as well as lower well down time in advance of oil production.

Once the nanocatalyst and/or carrier fluid has been injected into the injector and/or producer wells, the formation is heated using injected steam under pressure for a period of time to elevate the temperature in the formation to the desired viscosity and/or hydrocracking temperature. The pressure employed to inject the nanocatalyst and/or steam is not critical so long as it does not exceed the pressure handling capability of the formation. That is, the pressure is insufficient to cause fracturing of the formation. The pressure that can be tolerated by the formation is typically checked by the driller before the nanocatalyst is injected and the procedures for evaluating the pressure required to frack are well known in the industry. Typically the time required to heat the formation to the desired temperature is from about 1 to about 4 months, preferably about 2 to 4 months. This heating creates a heat chamber, within which the nanocatalyst is contained, preferably substantially at or in close proximity to the heat chamber's core. The heat chamber may extend 2, 3, 5, 10, or even 15 feet or more (in a radial direction from the perforated injector and/or producer well portions).

While the heat alone may reduce the viscosity of the heavy oil so that it flows by gravity, the nanocatalyst, at least in part by reducing the asphaltene cracking temperature, may further reduce the viscosity by initiating asphaltene hydrocracking and/or catalyzing asphaltene hydrocracking below the typical coking conditions. Other components in the heavy oil may also be hydrocracked down-well in the presence of the nanocatalysts described herein throughout. By non-limiting example, the viscosity of the contained oil in the heavy oil/extra heavy oil formation may range from about 100,000 to about 800,000 cps when measured at 60° F. (and all combinations and subcombinations of ranges thereof). After the well has been treated with the nanocatalyst using the methods disclosed herein for heavy oil wells, such as for example SAGD wells, the overall viscosity of the oil after removal from the well may be reduced to a range as low as, for example, from about 10,000 to about 80,000 cps (and all combinations and subcombinations of ranges thereof), when measured at 60° F. In other embodiments, the viscosity of the recovered oil after treatment and extraction from the well, when measured at 60° F., may be about 20% less, preferably about 30%, more preferably 40%, still more

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preferably 50, 60, 70, 80, 90, or even 95% less than the that of the oil contained in the well prior to treatment (when measured at 60° F.).

When the heavy oil formation, or at least the heat chamber formed within the formation by such steam heating, reaches the desired temperature, the steam is stopped in the one or more producer wells, which are then employed to transfer the now less viscous heavy oil from the formation to the surface. Steam may continue to be injected under pressure into the injection well (artificial pump relief), preferably is injected, but typically only to make up any pressure loss created in the formation as heavy oil (typically in combination with water from the previously injected steam) is removed and pumped to the surface via the producer well(s) and/or is needed to maintain the temperature in the heat chamber, where the oil and water may be separated.

The oil produced by such methods is typically less viscous, easier to extract from the well, and/or easier to process to final products than heavy oil extracted from wells without such treatment. It typically contains less asphaltenes and/or more products from asphaltene hydrocracking than heavy oil extracted from wells without such treatment. In many instances, more of the oil may be removed from the formation than without such treatment, and/or at lowered overall costs relative to untreated wells. In some embodiments, the extra-heavy oil may be converted to a heavy oil, accompanied by a change in physical properties associated with the two oils, such as viscosity and density relative to water.

The heavy oil found in these types for formations typically has an API gravity below about 22.3°. In extra heavy oil wells, the API gravity is typically below 10.0°. The heavy and/or extra heavy oil contains a number of components including asphaltenes as is readily recognized by the ordinarily skilled artisan.

In certain embodiments, the invention is directed to methods for upgrading heavy oil, preferably in a well, comprising:

contacting the heavy oil, preferably in a well that produces heavy oil, with a catalyst according to the invention for a time and under conditions sufficient to decrease the API gravity of the heavy oil. In certain of these embodiments, the initial API gravity of the heavy oil is between 8 and 10°. In other embodiments, the initial API gravity of the heavy oil is between 8 and 14. In other embodiments, the API gravity of the oil after treatment by a method as disclosed herein is reduced (i.e., the API gravity number is larger). In other embodiments, the initial API gravity of the heavy oil contained in the well is less than that of water (i.e., the oil sinks in water because the oil's density is heavier than water). In yet other embodiments, the API gravity of the heavy oil after treatment and extraction from the well is greater than that of water (i.e., the oil floats on water because the oil's density is lighter than water). In still other embodiments, the initial API gravity of the heavy oil contained in the well is less than that of water and the API gravity of the heavy oil after treatment and extraction from the well is greater than that of water. When the initial API gravity is between about 8.0° to about 9.5°, preferably the API gravity of the heavy oil after treatment and extraction from the well is 10.0 or higher, such as for example, a number up to about 22.3° (the API gravity range for heavy oil is 10 to) 22.3°, and all combinations and subcombinations of the range. When the initial API gravity is greater than 10°, the API gravity of the treated oil after extraction is typically in

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a range of from 12 to about 22.3°. In some instances, the API gravity of the heavy oil after treatment and extraction from the well may be even higher. After certain treatments, the extracted heavy oil typically has an API gravity in the range of from about 14.0° to about 22.3°.

The upgrading down well may take place in the following fashion. This disclosure is in part directed to employment of any of the methods disclosed herein in an SAGD well (horizontal producer and injector well legs, with producer leg(s) below the injector leg) formation containing heavy and/or extra heavy oil, but the procedure may also be satisfactorily employed in vertical steam-assisted wells. This general upgrading procedure may be employed after a well has been drilled and completed, whether or not the well is currently in production. Prior to introducing the catalyst into the well, it is useful if the well legs (injector and/or producer) are perforated (or contain slits) within the target zones that contain oil. A volume of treatment fluid containing catalyst is calculated, based on a radial volume of usually 6" to 1' surrounding the well bore in the target zone. This volume is calculated based on the effective pore volume attributable to the rock reservoir's porosity. The nanocatalyst/carrier fluid is transferred to the wellheads of at least some of injector/producer wells and a steam-assist is used to inject the nanocatalyst and carrier fluid into the well. The injection is carried out at a pressure higher than the formation static pressure, but less than its fracking pressure. Target formations (called pay zones or target zones) absorb the fluid as it is being injected. The pumping rate is set so as not to reach or exceed the formation fracture pressure, a characteristic defined by the geology of the individual well. Once the volume of the fluid has been squeezed from the well leg into the formation, injection ceases and the well is maintained for a period of time in a static condition (no oil removal), as disclosed herein, to allow the desired contacting between the nanocatalyst and the heavy/extra heavy oil to take place. During this time, the catalyst is in contact with the crude oil in the formation at the temperature and pressure that are defined by the well itself, the steam pressure and temperature and heat transfer efficiencies of the steam, the nanocatalyst and carrier fluids, and the contained oil, as well as the formation heat transfer coefficients. After sufficient time has been allowed for the contacting, the well is reopened (steam pressure removed from producer leg(s) and fluids (oil/water) from the target zones (pay Zone) begin to flow back to the surface via the producer leg(s) and the temperature and pressure in the formation is maintained with the assistance of steam under pressure.

In certain more preferred embodiments, the invention is directed to upgraded heavy oil fractions, preferably those produced in a heavy oil producing well, such fractions produced by the methods of the present invention. In certain of these embodiments, the upgraded heavy oil extracted from the well has an API gravity higher than the API gravity of the heavy oil in the formation prior to treatment of the heavy oil formation by the present methods.

In other embodiments, the nanocatalyst used to treat the heavy oil well has useful heat transfer properties and facilitates heat transfer from the injected steam into the formation. In certain embodiments, the use of the disclosed nanocatalysts reduces the amount of steam required to increase the temperature of the formation to the desired temperature for extraction of the heavy oil from the well and/or hydrocracking of contained asphaltenes.

In some other embodiments, use of the disclosed nanocatalysts in treatments described herein reduces or inhibits,

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preferably substantially reduces or substantially inhibits, more preferably substantially eliminates coking of heavy oil components within the formation during the time that the temperature is maintained in the well's heat chamber. In yet other embodiments, use of the disclosed nanocatalysts in treatments described herein enables the user to reduce the desired target temperature for the well's heat chamber, preferably while, maintaining, preferably increasing the amount of oil later extracted from the well, and/or facilitating the extraction of the oil.

In certain embodiments, the invention relates in part to methods for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to increase the API gravity of the heavy oil recovered from the well as compared to the API gravity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

In certain preferred embodiments, the nanocatalysts used in the above noted methods are steam-injected into the well producer and injector legs. In some embodiments, the producer and injector legs of said well are substantially parallel to each other and positioned substantially horizontally within the rock formation, with said producer leg positioned below the injector leg in the rock formation.

In other embodiments, the nanocatalysts used in the above noted methods further comprise a carrier fluid.

In yet other embodiments, a portion of the rock formation is steam-heated to a temperature in the range of from about 300° C. to about 500° C.

In still other embodiments, the rock formation temperature is maintained at the desired temperature (exemplary temperature in the range of from about 300° C. to about 500° C.) for a period of from about 2 months to about 4 months before the steam injection is discontinued in the producer leg of the well.

In certain other embodiments, after steam injection is discontinued in the producer leg of the well, the oil is extracted from the rock formation.

In certain other embodiments, the rock formation comprises oil sands containing the heavy oil or extra heavy oil.

In still other embodiments, the nanocatalyst further comprises a group VIII metal, preferably Pd or Pt.

In certain embodiments, the methods of treating heavy oil wells, such as those described herein throughout, are directed to viscosity-improved heavy oils or API-gravity increased heavy oils prepared by the methods of herein disclosed.

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In other embodiments, the invention is related in part to methods for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to decrease the viscosity of the heavy oil recovered from the well as compared to the viscosity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

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In still other embodiments, the rock formation temperature is maintained at the desired temperature (exemplary temperature in the range of from about 300° C. to about 500° C.) for a period of from about 2 months to about 4 months before the steam injection is discontinued in the producer leg of the well.

In certain other embodiments, after steam injection is discontinued in the producer leg of the well, the oil is extracted from the rock formation.

In certain other embodiments, the rock formation comprises oil sands containing the heavy oil or extra heavy oil.

In still other embodiments, the nanocatalyst further comprises a group VIII metal, preferably Pd or Pt.

In certain embodiments, the methods of treating heavy oil wells, such as those described herein throughout, are directed to viscosity-improved heavy oils or API-gravity increased heavy oils prepared by the methods of herein disclosed.

In other embodiments, the invention is directed to methods for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to increase the API gravity of the heavy oil recovered from the well as compared to the API gravity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the SBET surface area is from about 17 to about 70 m²/g.

In certain preferred embodiments, the nanocatalysts used in the above noted methods are steam-injected into the well producer and injector legs. In some embodiments, the producer and injector legs of said well are substantially parallel to each other and positioned substantially horizontally within the rock formation, with said producer leg positioned below the injector leg in the rock formation.

In other embodiments, the nanocatalysts used in the above noted methods further comprise a carrier fluid.

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In still other embodiments, the rock formation temperature is maintained at the desired temperature (exemplary temperature in the range of from about 300° C. to about 500° C.) for a period of from about 2 months to about 4 months before the steam injection is discontinued in the producer leg of the well.

In certain other embodiments, after steam injection is discontinued in the producer leg of the well, the oil is extracted from the rock formation.

In certain other embodiments, the rock formation comprises oil sands containing the heavy oil or extra heavy oil.

In still other embodiments, the nanocatalyst further comprises a group VIII metal, preferably Pd or Pt.

In certain embodiments, the methods of treating heavy oil wells, such as those described herein throughout, are directed to viscosity-improved heavy oils or API-gravity increased heavy oils prepared by the methods of herein disclosed.

In yet other embodiments, the invention is directed to methods for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to decrease the viscosity of the heavy oil recovered from the well as compared to the viscosity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the SBET surface area is from about 17 to about 70 m²/g.

In some preferred embodiments, the nanocatalysts used in the above noted methods are steam-injected into the well producer and injector legs. In some embodiments, the producer and injector legs of said well are substantially parallel to each other and positioned substantially horizontally within the rock formation, with said producer leg positioned below the injector leg in the rock formation.

In other embodiments, the nanocatalysts used in the above noted methods further comprise a carrier fluid.

In yet other embodiments, a portion of the rock formation is steam-heated to a temperature in the range of from about 300° C. to about 500° C.

In still other embodiments, the rock formation temperature is maintained at the desired temperature (exemplary temperature in the range of from about 300° C. to about 500° C.) for a period of from about 2 months to about 4 months before the steam injection is discontinued in the producer leg of the well.

In certain other embodiments, after steam injection is discontinued in the producer leg of the well, the oil is extracted from the rock formation.

In certain other embodiments, the rock formation comprises oil sands containing the heavy oil or extra heavy oil.

In still other embodiments, the nanocatalyst further comprises a group VIII metal, preferably Pd or Pt.

In certain embodiments, the methods of treating heavy oil wells, such as those described herein throughout, are directed to viscosity-improved heavy oils or API-gravity increased heavy oils prepared by the methods of herein disclosed.

Once armed with the disclosures provided herein, the skilled artisan will be able to appreciate and employ to great advantage the methods, techniques and nanocatalysts disclosed herein for the treatment of heavy oil containing wells, preferably SAGD-type heavy oil wells.

The disclosures of each of the foregoing documents are hereby incorporated herein by reference, in their entireties.

The present invention is further described in the following examples. Excepted where specifically noted, the examples are actual examples. These examples are for illustrative purposes only, and are not to be construed as limiting the appended claims.

EXPERIMENTAL SECTION

Examples of the Present Invention

Example 1 (a) Synthesis of Alumina Nanoparticles

An alumina useful in making a catalyst of the present invention was provided in the following manner. Commercially available pure aluminum powder (54 g, 99.2%Al) was dissolved in 270 mL of 50% sodium hydroxide solution at 92° C. After the aluminum dissolved, the solution was allowed to cool to room temperature and filtered. The sodium aluminate was slowly precipitated from solution by adding 30 g of ethyl alcohol and 100 g of distilled water to the solution with gentle stirring. Gaseous CO₂ was slowly bubbled into the solution at room temperature with gentle stirring while the pH was monitored. After 40 minutes, the solution reached a pH of approximately 8.5. At this point, the introduction of gaseous CO₂ was terminated. The resulting precipitate was separated from the mother liquor by filtration and rinsed two times with distilled water until the washes attained a pH 7.0-7.5. The washed sodium aluminate

precipitate was dried by heating for 90 minutes in a muffle furnace at 120° C. ($\text{Na}_2\text{O}/\text{Al}_2\text{O}_3$ ratio of sodium aluminate obtained was lower than 1.2).

(b) Alternate Synthesis of Alumina Nanoparticles

An alumina useful in making a catalyst of the present invention was provided in the following manner. Commercially available aluminum hydroxide powder (Purity 99.5%, 158 g.) was dissolved in 270 mL of 30% sodium hydroxide solution at 92° C. After the aluminum dissolved, the solution was allowed to cool to room temperature and filtered. The sodium aluminate was slowly precipitated from solution by adding 30 g of ethyl alcohol and 100 g of distilled water to the solution with gentle stirring. Gaseous CO_2 was slowly bubbled into the solution at room temperature with gentle stirring while the pH was monitored. After 40 minutes, the solution reached a pH of approximately 8.5. At this point, the introduction of gaseous CO_2 was terminated. The resulting precipitate was separated from the mother liquor by filtration and rinsed two times with distilled water until the washes attained a pH 7.0-7.5. The washed sodium aluminate precipitate was dried by heating for 90 minutes in a muffle furnace at 120° C. ($\text{Na}_2\text{O}/\text{Al}_2\text{O}_3$ ratio of sodium aluminate obtained was lower than 1.2).

(c) Synthesis of Supported Nickel Oxide Nanoparticles on Alumina Nanoparticles

The dried sodium aluminate precipitate (100 g) from Step (a) ["alumina source"] was impregnated with 4 grams of a 5% by weight aqueous nickel nitrate $\text{Ni}(\text{NO}_3)_2$ solution for 3 hours using the incipient wetness technique. The nickel wetted precipitate was dried at 120° C. for 6 hours and then calcinated at 450° C. for 6 hours. Samples of calcinated product (supported nanocatalyst) were characterized by N_2 adsorption at -196° C. and X-ray diffraction (XRD). Nitrogen adsorption isotherms were obtained with an Autosorb-1 from Quantacrome after outgassing samples overnight at 140° C. under high vacuum (10^{-6} mbar). Surface area (S_{BET}) values were calculated using the model of Brunauer, Emmet and Teller (BET). X-Ray Diffraction patterns were recorded with a Philips PW1710 diffractometer using $\text{Cu K}\alpha$ radiation to characterize the catalyst and measure particle size. Results are shown in Table 1.

TABLE 1

Surface Characteristics of Support Nanoparticles and Supported Nickel Oxide on Alumina Nanoparticles			
Material	$S_{\text{BET}}(\text{m}^2/\text{g})$	$d_{\text{p-Al}}(\text{nm})$	$d_{\text{p-NiO}}(\text{nm})$
Alumina (Al)	123.2	35 ± 4	—
AlNi (from 5% sol.)	69.9	35 ± 4	16
AlNi (from 15% sol.)	17.9	35 ± 4	29

(d) The dried sodium aluminate precipitate (100 g) from Step (a) ["alumina source"] was impregnated with 4 grams of a 15% by weight aqueous nickel nitrate $\text{Ni}(\text{NO}_3)_2$ solution for 3 hours using the incipient wetness technique. The nickel wetted precipitate was dried at 120° C. for 6 hours and then calcinated at 450° C. for 6 hours. Samples of calcinated product (supported nanocatalyst) were characterized by N_2 adsorption at -196° C. and X-ray diffraction (XRD). Nitrogen adsorption isotherms were obtained with an Autosorb-1 from Quantacrome after outgassing samples overnight at 140° C. under high vacuum (10^{-6} mbar). Surface area (S_{BET})

values were calculated using the model of Brunauer, Emmet and Teller (BET). X-Ray Diffraction patterns were recorded with a Philips PW1710 diffractometer using $\text{Cu K}\alpha$ radiation to characterize the catalyst and measure particle size. Results are shown in Table 1.

(e) Asphaltenes adsorption experiments. A calibration curve of UV absorbance versus asphaltene concentration at 400 nm was constructed from the prepared solutions with known concentrations. Toluene was used as solvent for dilution of the asphaltenes, and for the blank in calibration curve construction. To asphaltene solutions of constant volume (10 mL) was added a constant amount (100 mg) of nanoparticles of the supported catalyst containing nickel oxide nanoparticles on alumina nanoparticles obtained in previous step (b) or (c) or a comparison sample containing alumina nanoparticles (100 mg) as described herein. A sample of each solution was stirred at 200 rpm for 10 hours at each of the following temperatures (25, 40, 55 and 70° C.) and analyzed in order to determine the equilibrium for sorption of the asphaltenes. Measurements were taken periodically to monitor progress toward sorption equilibrium. The results are shown in FIGS. 1 to 4. Sorption analysis of asphaltene concentration measurements indicated that one hour was sufficient time to achieve the thermodynamic sorption equilibrium. To determine kinetic parameters, the amount of asphaltenes adsorbed was calculated from concentration measurements for a range of asphaltene concentrations (250, 750, 1500 and 2000 mg/L initial concentrations) at different times as shown in FIGS. 5 to 6. The results indicate that equilibrium is reached more quickly at lower initial concentrations (2 min for calcinated Alumina, AlNi 5% nanocatalyst¹, and AlNi 15% nanocatalyst² at 250 and 750 mg/L asphaltene initial concentrations). In contrast, equilibrium is attained after 80 minutes for higher initial asphaltene concentrations (80 minutes for AlNi 5% nanocatalyst, and AlNi 15% nanocatalyst at 1500 and 2000 mg/L asphaltene initial concentrations, respectively).

Example 2

Preparation of dried sodium aluminate precipitate with low sodium content was carried out as in the Example 1 Step (a). The dried sodium aluminate precipitate (100 g) from Step (a) ["alumina source"] was impregnated with 3 grams of a 5% by weight aqueous nickel nitrate $\text{Ni}(\text{NO}_3)_2$ solution for 3 hours using the incipient wetness technique. The nickel wetted precipitate was dried at 120° C. for 6 hours. Dried sodium aluminate precipitate impregnated with nickel salt (100 g) was impregnated with 0.5 g of a 2% by weight aqueous Palladium nitrate $\text{Pd}(\text{NO}_3)_2$ solution for 3 hours using the incipient wetness technique. The palladium wetted precipitate was dried at 120° C. for 6 hours and then calcinated at 550° C. for 6 hours. Samples of calcinated bimetallic oxide product (supported nanocatalyst referred as Pd0.5/Ni5/Al) were characterized by N_2 adsorption at -196° C. and X-ray diffraction (XRD). Nitrogen adsorption isotherms were obtained with an Autosorb-1 from Quantacrome after outgassing samples overnight at 140° C. under high vacuum (10^{-6} mbar). Surface area (S_{BET}) values were calculated using the model of Brunauer, Emmet and Teller (BET). X-Ray Diffraction patterns were recorded with a Philips PW1710 diffractometer using $\text{Cu K}\alpha$ radiation to characterize the catalyst and measure particle size such size was 35 nm. Results of kinetics and isotherm analyses are shown below:

Example 3

Heavy oil from the San Vicente oilfield in Colombia was upgraded in the following manner. In a 1500 mL stainless-

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steel batch reactor 200 g extra-heavy oil, 67 mL prepared 1% wt. aqueous sodium chloride and nanocatalyst (5000 ppm of the supported catalyst containing nickel oxide nanoparticles on alumina nanoparticles obtained in Example 1 (labeled as AlNi (15%)), the 5000 ppm based on the weight of heavy oil. The air above the reaction mixture was replaced with water steam in order to evacuate the air in headspace head of the vessel. The reaction mixture was heated to 300° C. and a relative pressure at temperature of 300 bar in the reactor. The temperature was maintained for 6 hours. The reaction mixture was cooled to room temperature, allowing the water and oil to separate. The water was drained from the bottom of the reactor. The oil remaining in the reactor was removed and analyzed for any upgrading. With respect to the original crude oil, the viscosity of the upgraded sample was reduced approximately 90% by the catalytic thermal cracking reaction. Additionally the API gravity of the original oil was improved from 8 API gravity degrees to 19 API gravity degrees. The API Gravity was measured by ASTM D287 (Hydrometer Method).

Example 4

Samples of asphaltenes from the crude oil were isolated by following a well known procedure. (See Kokal, S. L., J. Najman, S. G. Sayegh, and A. E. George, "Measurement and Correlation of Asphaltene Precipitation from Heavy Oils by Gas Injection," *J. Can. Petrol. Technol.*, 31, 24 (1992). An excess of n-heptane (99% Sigma Aldrich) was added to the crude oil from La Hocha oilfield in Colombia in a volume ratio of 40:1. The mixture was sonicated for 2 hours at 25° C. and further stirred at 300 rpm for 20 hours. The precipitated fraction (10%) was isolated by filtration using a 8 µm Whatman filter paper and washed with n-heptane at a ratio of 4/1 (g/mL). The precipitated fraction containing asphaltenes was added to n-heptane, and the mixture was centrifuged at 5000 rpm for 15 minutes and left to rest for 24 hours. The cake was washed with n-heptane several times until the color of the asphaltenes became shiny black. The sample was dried in a vacuum oven at 25° C. for 12 hours. The obtained asphaltene sample was homogenized in a mortar. The homogenized asphaltenes were dissolved in toluene for preparing a stock solution at 3000 mg/L. Solutions with different concentrations (150, 250, 400, 750, 1000, 1500 and 2000 mg/L) were prepared from the stock solution. The asphaltene adsorption test to evaluate adsorption on Nickel oxide nanoparticles supported on alumina nanoparticles was performed following same protocol described in Example 1, step (c).

Catalytic steam gasification of adsorbed asphaltenes on nanoparticle catalysts was evaluated using a simultaneous thermogravimetric analysis/differential scanning calorimetry (TGA/DSC) analyzer (SDT Q600, TA Instruments, Inc., New Castle, DE). The instrument had a horizontal beam design that allowed flow of gas parallel to the beam as well as above the sample. The system was also equipped with an outlet close to the sample for steam injection. A sample of each nanocatalyst (similarly prepared to the procedure outlined in Example 1) having approximately 10 mg asphaltenes adsorbed thereon was tested. The same procedure was performed with 10 mg of pure asphaltenes for comparison. The amount of sample employed was chosen to avoid diffusion limitations. Gasification was performed by first purging the system with argon (Ar) at a flow rate of 500 cm³/min for 10 minutes, then decreasing the flow rate to 100 cm³/min, and maintaining this flow throughout the experiment. After the system was purged with argon (Ar) at a flow

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rate to 100 cm³/min for 20 minutes at room temperature, the temperature was abruptly raised to 150° C. At the same time, H₂O(g) was introduced to the system at a flow rate of 6.30 cm³/min. This flow rate allowed the steam to be present above the sample in excess. The temperature of the system was increased at a of rate 5° C. per minute until a temperature of 800° C. was achieved, while recording mass changes in the sample using thermogravimetric analysis/differential scanning calorimetry (TGA/DSC) analyzer (SDT Q600, TA Instruments, Inc., New Castle, DE). The mass changes are indicative of the initiation and propagation of the cracking gasification reaction.

Thermal catalytic cracking of asphaltenes extracted from extra heavy oil in presence of nickel oxide nanoparticles catalyst supported on alumina nanoparticles was shown to be effective. As a result, with respect to the original asphaltenes, the catalytic cracking temperature of the asphaltenes in the presence of the supported catalyst containing nickel oxide nanoparticles on alumina nanoparticles obtained in Example 1c (AlNi(5%)) and Example 1d (AlNi(15%)) was approximately 300° C. and 220° C., respectively, as compared with a cracking temperature for asphaltenes of approximately 540° C. in the absence of catalyst. Table 2 show cracking temperatures and the enhancement observed with the supported nanoparticles catalyst.

TABLE 2

Cracking Temperatures of Asphaltenes	
Material	T (° C.)-Cracking
Asphaltene (A)	540
Asphaltenes + Alumina	520
Asphaltenes + AlNi5	300
Asphaltenes + AlNi15	220

Example 5

Upgrading of a Heavy Oil with a Bimetallic Oxide Supported Nanocatalyst

Heavy Oil (200 g) from Chichimene-17 (a Colombian oil well producing very heavy oil) was placed into a high pressure stirred autoclave reactor. The Pd0.5/Ni5/Al bimetallic oxide nanocatalyst obtained from Example 2 (5 g) was added with vigorous stirring. The hydrogen donor 1,2,3,4, Tetrahydronaphthalene (20 g, from Merck Chemicals) was placed into the reactor and the reactor was heated to 80° C. with stirring. When the internal temperature of the reactor reached 80° C., the system was pressurized to 1500 psi with hydrogen gas and maintained at this temperature and pressure continued for 6 hours at 80° C. The reaction mixture was allowed to stand at room temperature without additional external cooling for four hours. The treated oil from the reactor was collected for analysis. Feedstock oil (100 g) from Chichimene-17 containing 1,2,3,4, Tetrahydronaphthalene (10 g) and the treated oil from the reactor were analyzed for H/C ratio in elemental analyzer EXETER CE-490. The results are shown in Table 3.

TABLE 3

Properties of Crude and Treated Heavy Oil Feedstock		
	Properties of Chichimene-17 Crude Oil Feedstock	Properties of Upgraded Oil
Carbon (wt %)	87.97	75.42
Hydrogen (wt %)	11.32	11.87
H/C ratio	1.53	1.87

The data support the conclusion that there is less unsaturation in the treated oil, indicating that the heavy oil has been upgraded at a moderate temperature (80° C. and without the necessity of using steam.

Example 6

To evaluate the Alumina supported Ni/Pd nanocatalyst (about 0.5% of Ni oxide nanoparticles and 0.1% of Pd oxide nanoparticles and prepared according to methods described herein) in a process of long term steam injection for hydrocracking of heavy fraction oil fractions, a sample of an extra-heavy oil (500 grams) having an API of 9.5 API as measured by API ASTM D287 was poured into the heater chamber as depicted in FIG. 1. Subsequently, the Ni/Pd nanocatalyst (0.5 grams) was added to the extra heavy oil sample. The heater chamber containing the heavy oil and Ni/Pd nanocatalyst was then connected to a steam generator and the catalyst/oil mixture allowed to heat for a period of 3 weeks at a temperature in a range of about 273° C. to about 345° C. and a steam pressure in a range of about 1300 Psi to about 2800 psi. After 3 weeks of exposure to steam heating in the chamber, the steam was discontinued, the chamber was allowed to cool over a period of two days, and then depressurized. An oil sample was collected from the heater chamber, the water was separated by centrifugation, and the API gravity of the separated oil was measured using method API ASTM D287, and the results are reported as shown in the following table:

Sample	API Gravity measured by API ASTM D287
Oil sample before treatment	9.5° API measured at 60° F.
Oil sample after nanocatalyst treatment and steam heating	17.0° API measured at 60° F.

The 17.0° API gravity measured for the final product sample is reflective of a reduction in overall viscosity as compared with the sample before treatment with nanocatalyst and extended steam treatment. The conditions chosen were employed to exemplify, in a non-limiting way, the kind of typical down-well conditions experienced in certain heavy well and/or heavy oil formations.

Example 7 (Hypothetical)

Heavy oil from an oilfield is upgraded in the following manner. At a time after a SAGD heavy oil well in the oilfield has been drilled and completed, including perforations to the injector and producer well legs to allow for steam introduction into the well target zones containing the oil, a volume of nanocatalyst fluid containing a Ni/Group VIII nanocatalyst appropriate for the well is calculated, based on a radial volume of usually 6" to 1' surrounding the well bore in the target zone and extending substantially along the length of the perforated section of the well leg. This volume is

calculated for the effective pore volume based on the rock reservoir porosity. The catalyst containing carrier fluid is injected into the well at the well head in each of the injector and producer well legs, with the assistance of steam pressure to drive the catalyst containing fluid down well and out into the oil through the well leg perforations in the target zone (pay zone) area of the reservoir. The pressure is set so as not to reach or exceed the particular formation's fracture pressure, a characteristic defined by the geology of the individual well. Once the volume of the fluid is injected into the formation, and the well is maintained for about 3 months in a static condition (no oil removal) while maintaining a steam pressure to elevate and hold the forming heat chamber at a temperature of about 350 to 450° C. to allow the desired contacting of the catalyst with the heavy oil to take place. During this entire time, the catalyst is in contact with the crude oil in the formation at the temperature and pressure that are defined by the well itself. After a period of about 3 months, the steam injection is stopped in the producer well leg(s), the well is reopened, oil begins to migrate in the direction of the producer well leg(s) under the force of gravity, and fluids (heavy oil/water) from the producer leg target zone(s) (pay Zone) begin to flow back to the surface. The API gravity of the produced oil is measured and is reflective of a reduction in overall viscosity as compared with a core sample taken before treatment with nanocatalyst and extended steam treatment.

Embodiment 1. A catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500.

Embodiment 2. A catalyst according to Embodiment 1, wherein the ratio is in a range of from about 99 to about 400.

Embodiment 3. A catalyst according to Embodiment 1 or 2, wherein the Nickel oxide (NiO) nanoparticles are present in an amount of about 0.2% to about 1% by weight of catalyst as measured by hydrogen adsorption.

Embodiment 4. A catalyst according to any one of Embodiments 1 to 3, wherein the particle size of the nickel oxide nanoparticles or the alumina nanoparticles is less than about 0.1 μm.

Embodiment 5. A catalyst according to Embodiment 4, wherein the particle size of the nickel oxide nanoparticles and the alumina nanoparticles are each less than about 0.1 μm.

Embodiment 6. A catalyst according to any one of Embodiments 1 to 5, wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

Embodiment 7. A catalyst according to any one of Embodiments 1 to 6, further comprising nanoparticles of at least one Group VIIIB metal oxide supported on the alumina nanoparticles;

wherein:

the Group VIIIB metal is other than nickel; and
the alumina nanoparticle to Group VIIIB metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500.

Embodiment 8. A catalyst according to any one of Embodiments 1 to 7, further comprising nanoparticles of at least one Group IB metal supported on the alumina nanoparticles;

wherein:

the alumina nanoparticle to Group IB metal nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500.

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Embodiment 9. A process for preparing a catalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500;

said process comprising

dry impregnating an amorphous dried sodium aluminate precipitate with an aqueous solution of a water-soluble nickel salt; and

drying the nickel impregnated precipitate;

wherein the dry impregnating and drying steps are each carried out for a time and under conditions sufficient to provide the dried nickel impregnated precipitate catalyst.

Embodiment 10. A process according to Embodiment 9, wherein the dried nickel impregnated precipitate is calcined in the presence of oxygen or air for a time and under conditions sufficient to provide the calcined catalyst.

Embodiment 11. A process according to Embodiment 9 or 10, wherein the nickel impregnated precipitate is dried at a temperature in the range of from about 100 to about 140° C. for from about 3 to about 8 hours.

Embodiment 12. A process according to Embodiment 10, wherein the dried nickel impregnated precipitate is calcined at a temperature in the range of from about 400 to about 500° C. for from about 3 to about 8 hours.

Embodiment 13. A process according to any one of Embodiments 9 to 12, wherein the nickel salt comprises nickel nitrate, nickel chloride or nickel sulfate.

Embodiment 14. A process according to Embodiment 13, wherein the nickel salt comprises nickel nitrate.

Embodiment 15. A process according to any one of Embodiments 9 to 14, wherein the catalyst further comprises nanoparticles of at least one Group VIII B metal oxide nanoparticles supported on the alumina nanoparticles;

wherein:

the Group VIII B metal is other than nickel; and

the alumina nanoparticle to Group VIII B metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500;

said process further comprising:

dry impregnating the dried nickel impregnated precipitate with an aqueous solution of a water-soluble Group VIII B metal salt;

drying the nickel and Group VIII B metal impregnated precipitate; and

calcining the dried nickel and Group VIII B metal impregnated precipitate in the presence of oxygen or air;

wherein:

each of the dry impregnating, drying, and calcining are carried out for a time and under conditions sufficient to provide the calcined catalyst.

Embodiment 16. A process according to any one of Embodiments 9 to 15, wherein the catalyst further comprises nanoparticles of at least one Group IB metal supported on the alumina nanoparticles;

wherein:

the alumina nanoparticle to Group IB metal nanoparticle weight to weight ratio in the catalyst is in a range of from about 80 to about 500;

said process further comprising:

dry impregnating the dried nickel impregnated precipitate with an aqueous solution of a water-soluble Group IB metal salt;

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drying the nickel and Group IB metal impregnated precipitate; and

calcining the dried nickel and Group IB metal impregnated precipitate in the presence of oxygen or air;

wherein:

each of the dry impregnating, drying, and calcining are carried out for a time and under conditions sufficient to provide the calcined catalyst.

Embodiment 17. A catalyst prepared by the process of any one of Embodiments 9 to 16.

Embodiment 18. A method for upgrading heavy oil fractions in a well, comprising:

contacting the heavy oil in a well producing heavy oil with a catalyst according to any one of Embodiments 1 to 8, for a time and under conditions sufficient to increase the H/C ratio.

Embodiment 19. A method according to Embodiment 18 further comprising a hydrogen transfer agent.

Embodiment 20. A method according to Embodiment 18, wherein the hydrogen transfer agent comprises 1,2,3,4-tetrahydronaphthalene, and wherein the method further optionally comprises the presence of hydrogen gas.

Embodiment 21. An upgraded heavy oil fraction prepared by the process of any one of Embodiments 18 to 20.

Embodiment 22. A method for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to increase the API gravity of the heavy oil recovered from the well as compared to the API gravity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

Embodiment 23. A method for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to decrease the viscosity of the heavy oil recovered from the well as compared to the viscosity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

Embodiment 24. A method for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to increase the API gravity of the heavy oil recovered from the well as compared to the API gravity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the SBET surface area is from about 17 to about 70 m²/g.

Embodiment 25. A method for upgrading heavy oil in a steam-assisted heavy oil well, comprising:

contacting the heavy oil contained in a rock formation associated with a steam-assisted well for producing heavy oil;

wherein said contacting of the heavy oil includes contacting with a nanocatalyst for a time and under conditions sufficient to decrease the viscosity of the heavy oil recovered from the well as compared to the viscosity of the heavy oil before said contacting;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

wherein the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the SBET surface area is from about 17 to about 70 m²/g.

Embodiment 26. A method according to any one of Embodiments 22 to 25, wherein said catalyst is steam-injected into the well producer and injector legs.

Embodiment 27. A method according to any one of Embodiments 22 to 26, wherein said catalyst further comprises a carrier fluid.

Embodiment 28. A method according to any one of Embodiments 26 to 27, wherein said producer and injector legs of said well are substantially parallel to each other and

positioned substantially horizontally within the rock formation, with said producer leg positioned below the injector leg in the rock formation.

Embodiment 29. A method according to any one of Embodiments 22 to 28, wherein a portion of the rock formation is steam-heated to a temperature in the range of from about 300 C to about 500 C.

Embodiment 30. A method according to any one of Embodiments 22 to 29, wherein the rock formation temperature is maintained at said temperature for a period of from about 2 months to about 4 months before the steam injection is discontinued in the producer leg of the well.

Embodiment 31. A method according to any one of Embodiments 22 to 30, wherein, after steam injection is discontinued in the producer leg of the well, the oil is extracted from the rock formation.

Embodiment 32. A method according to any one of Embodiments 22 to 31, wherein the rock formation comprises oil sands containing the heavy oil or extra heavy oil.

Embodiment 33. A method according to any one of Embodiments 22 to 32, wherein the nanocatalyst further comprises a group VIII metal.

Embodiment 34. A viscosity-improved heavy oil prepared by the method of any one of Embodiments 22 to 33.

Embodiment 35. An API-gravity increased heavy oil prepared by the method of any one of Embodiments 22 to 34.

When any variable occurs more than one time in any constituent or in any formula, its definition in each occurrence is independent of its definition at every other occurrence. Combinations of substituents and/or variables are permissible only if such combinations result in stable compositions.

It is believed the chemical formulas, abbreviations, and names used herein correctly and accurately reflect the underlying compounds reagents and/or moieties. However, the nature and value of the present invention does not depend upon the theoretical correctness of these formulae, in whole or in part. Thus it is understood that the formulas used herein, as well as the chemical names and/or abbreviations attributed to the correspondingly indicated compounds, are not intended to limit the invention in any way, including restricting it to any specific form or to any specific isomer.

When ranges are used herein for physical properties, such as molecular weight, API gravity, viscosity, surface area, particle size, or chemical properties, such as chemical formulae, contacting times of reagents, drying and calcining times, pressures and temperatures, all combinations and subcombinations of ranges and specific embodiments therein are intended to be included.

The disclosures of each patent, patent application and publication cited or described in this document are hereby incorporated herein by reference, in their entirety.

The invention illustratively disclosed herein suitably may be practiced in the absence of any element which is not specifically disclosed herein. The invention illustratively disclosed herein suitably may also be practiced in the absence of any element which is not specifically disclosed herein and that does not materially affect the basic and novel characteristics of the claimed invention.

Those skilled in the art will appreciate that numerous changes and modifications can be made to the preferred embodiments of the invention and that such changes and modifications can be made without departing from the spirit of the invention. It is, therefore, intended that the appended claims cover all such equivalent variations as fall within the true spirit and scope of the invention.

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What is claimed:

1. A method for upgrading a heavy oil fraction in an oil well comprising:

contacting a heavy oil in a rock formation associated with the well producing the heavy oil with a nanocatalyst for a time and under conditions sufficient for the nanocatalyst to adsorb an asphaltene in the heavy oil;

wherein:

the heavy oil comprises the asphaltene;

the heavy oil fraction upgrading comprises adsorbing the asphaltene in the heavy oil onto the nanocatalyst in the well;

said nanocatalyst comprises:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein:

the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst or the SBET surface area is from about 17 to about 70 m²/g; and

subsequent to the contacting, the well is maintained in a static condition for a period of time before heavy oil removal from the well is initiated.

2. A method according to claim 1, wherein the contacting is carried out under intrinsic well temperature conditions.

3. A method according to claim 1, wherein the contacting is carried out under steam-assisted well conditions.

4. A method according to claim 3, wherein the well comprises a steam-assisted gravity drainage (SAGD) well.

5. A method according to claim 1, wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

6. A method according to claim 1, wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 400.

7. A method according to claim 1, wherein the catalyst further comprises a carrier fluid.

8. A method according to claim 1, wherein the catalyst further comprises Group VIII metal oxide nanoparticles supported on alumina nanoparticles, wherein the Group VIII metal oxide nanoparticles are other than nickel nanoparticles;

wherein the alumina nanoparticle to Group VIII metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 400;

wherein the particle size of the alumina nanoparticle is in a range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of the catalyst.

9. A method according to claim 1, wherein the contacting does not exceed the intrinsic fracture pressure of the oil well.

10. A method according to claim 1, wherein, after the static condition, the heavy oil in the well is retreated by contacting with a nanocatalyst for a time and under conditions sufficient for the nanocatalyst to adsorb a further amount of the asphaltene;

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said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein:

the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst or the SBET surface area is from about 17 to about 70 m²/g.

11. A method according to claim 3, wherein the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst.

12. A method according to claim 3, wherein the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 400.

13. A method according to claim 3, wherein the catalyst further comprises a carrier fluid.

14. A method according to claim 3, wherein the catalyst further comprises Group VIII metal oxide nanoparticles supported on alumina nanoparticles, wherein the Group VIII metal oxide nanoparticles are other than nickel nanoparticles;

wherein the alumina nanoparticle to Group VIII metal oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 400;

wherein the particle size of the alumina nanoparticle is in a range of from about 30 to about 100 nanometers;

wherein the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

wherein the alumina nanoparticles are present in an amount of at least 99% by weight of the catalyst.

15. A method according to claim 3, wherein the contacting does not exceed the intrinsic fracture pressure of the oil well.

16. A method according to claim 3, wherein the heavy oil in the well is retreated by contacting with a nanocatalyst for a time and under conditions sufficient for the nanocatalyst to adsorb the asphaltene;

said nanocatalyst comprising:

nickel oxide nanoparticles supported on alumina nanoparticles;

wherein:

the alumina nanoparticle to nickel oxide nanoparticle weight to weight ratio in the catalyst is in a range of from about 99 to about 500;

the particle size of the alumina nanoparticle is in the range of from about 30 to about 100 nanometers;

the catalyst does not further comprise silver nanoparticles supported on the alumina nanoparticles; and

the alumina nanoparticles are present in an amount of at least 99% by weight of catalyst or the SBET surface area is from about 17 to about 70 m²/g.